

Multiple-relaxation-time lattice Boltzmann method for the Navier-Stokes and nonlinear convection-diffusion equations: Modeling, analysis, and elements

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In this paper, we first present a unified framework of multiple-relaxation-time lattice Boltzmann (MRT-LB) method for the Navier-Stokes and nonlinear convection-diffusion equations where a block-lower-triangular-relaxation matrix and an auxiliary source distribution function are introduced. We then conduct a comparison of the four popular analysis methods (Chapman-Enskog analysis, Maxwell iteration, direct Taylor expansion, and recurrence equations approaches) that have been used to obtain the macroscopic Navier-Stokes and nonlinear convection-diffusion equations from the MRT-LB method and show that from mathematical point of view, these four analysis methods can give the same equations at the second-order of expansion parameters. Finally, we give some elements that are needed in the implementation of the MRT-LB method and also find that some available LB models can be obtained from this MRT-LB method.

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I. INTRODUCTION

Fluid flows, heat, and mass transfer are the most common basic phenomena in nature, science, and engineering, and usually they can be described by the classical Navier-Stokes equations (NSEs) and nonlinear convection-diffusion equations (NCDEs) [1]. However, it is difficult to obtain their analytical solutions due to the coupling between the nonlinear NSEs and NCDEs. With the development of computing technique, the numerical simulation has been becoming more popular in the study of the complex fluid flows, heat, and mass transfer. The lattice Boltzmann (LB) method, as one of mesoscopic numerical approaches, has attained increasing attention and is very successful in the simulation of the complex physical systems for its distinct advantages in kinetic background, treatment of complex boundary conditions, and computational efficiency in parallel systems [2–5].

Based on the collision term, there are three popular LB models [4,5]. The first one is the lattice Bhatnagar-Gross-Krook (BGK) model in which only one relaxation time is introduced into the collision term, and hence it is also usually called the single-relaxation-time LB (SRT-LB) model. Historically, this model, as one of the most popular LB models for its simplicity and efficiency, has been proposed independently by two different research groups [6,7], and later it was also found that the SRT-LB model can be considered a special discretized form of the continuous BGK-Boltzmann equation [8]. Almost at the same time, based on the previous work on collision operator [9–11], the second one, named multiple-relaxation-time or generalized LB (MRT-LB) model,

was developed by d’Humières [12]. Compared to the SRT-LB model, the collision of the MRT-LB model is depicted with a matrix that allows for a decoupled relaxation of different moments and includes all possible degrees of freedom that can be used to optimize the LB method in terms of stability and accuracy [13–18]. The third one is the two-relaxation-time LB (TRT-LB) model in which the collision term (or nonequilibrium distribution function) is decomposed of symmetric (even) and antisymmetric (odd) parts, which is first developed by Ginzburg [19–22]. This model, through introducing two different relaxation parameters ($s^+ = 1/\tau^+$ and $s^- = 1/\tau^-$ with τ^+ and τ^- being relaxation times corresponding to the symmetric and antisymmetric modes, they are also two eigenvalues of the collision matrix) in the collision term, possesses the flexibility of the MRT-LB model and the simplicity of SRT-LB model and also retains the computational efficiency [16,23]. Actually, besides the Ginzburg’s TRT-LB model [19–21], the lattice kinetic (LK) scheme [24] (the mass conservation cannot be satisfied), the modified lattice kinetic (MLK) schemes [25–27] with two different relaxation parameters $1/\tau$ and $1/(\tau - A)$ (A is parameter that can be used to adjust the value of relaxation time τ), and the regularized LB (RLB) model [28–32] with two special relaxation parameters $1/\tau$ and 1 can also be viewed as some special TRT-LB models [23,33], but they are different from the one developed by Ginzburg [19–21] in the collision matrix. In addition, to improve the stability and accuracy of LB method, some recursive regularized LB models were also developed [34–36], while these models cannot be written as the TRT-LB models [37]. Meng and Guo [38] also developed another version of the MRT-LB model which combines the commonly used MRT-LB and MLK schemes [12,13,24–26] and found that compared to the popular MRT-LB model [12,13], their model is more stable. Recently, Zhao *et al.* [39] proposed a block

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triple-relaxation-time LB model for NCDEs in which a block diagonal relaxation matrix composed of three blocks is introduced. They also found that the above-mentioned RLB models and the MLK schemes are some particular forms of the block triple-relaxation-time LB model. However, one can show that the model developed by Zhao *et al.* [39] is also a particular MRT-LB version. In addition to the above-mentioned LB models, to improve the stability of the SRT-LB model, some different LB models are also developed. For instance, Karlin and his collaborators [40–43] proposed the entropic LB models where the relaxation time and equilibrium distribution function are given properly to satisfy the discrete H theorem, while these models can also be considered as the special SRT versions. Then under the assumption of fast-slow decomposition of motions near the quasiequilibrium manifold, they also developed the two-relaxation-time entropic LB models for multicomponent fluid flows [44–46]. Geier *et al.* [47] developed a cascaded or central-moment LB model where a natural way is presented to maintain Galilean invariance for a given velocity set, which can also be realized through the use of appropriate correction terms and a guided equilibrium model [48]. Actually, the central-moment LB model is also a particular version of the MRT-LB model where the collision matrix is related to macroscopic velocity [37,49,50]. Finally, we also point out that in the SRT-LB, entropic LB, TRT-LB, RLB, MLK, and block triple-relaxation-time LB models, the collision process is conducted in velocity space. However, the collision processes of the MRT-LB and central-moment LB models are usually implemented in momentum space.

As discussed above, although many different LB models have been developed in the past decades, most of them can be viewed as special forms of the MRT-LB model (the cumulant LB model [51] cannot be written in a form of the MRT-LB model since the transformation between the discrete velocity and moment space is nonlinear [37]). It should also be noted that as a more general version, however, the MRT-LB model still has some issues needed to be addressed. For instance, it is unclear how to choose the eigenvectors of the collision matrix in the MRT-LB model besides the ones corresponding to the conserved variables (e.g., the density and momentum appeared in the NSEs); this is mainly caused by their nonuniqueness. In addition, the analysis of MRT-LB model is usually limited to the specified dimensional space and lattice structure (e.g., D2Q9 lattice [4,12,17]), and compared to the SRT-LB model, it seems more difficult to present a general analysis to derive the macroscopic governing equations. Here we point out that Adhikari and Succi have proposed a guideline to choose the eigenvectors through introducing the notion of duality [52]. Furthermore, based on the Taylor expansion, Kaehler and Wagner have performed a general analysis (not restricted to a specific lattice structure) to the MRT-LB model and also gave some necessary requirements on the equilibrium distribution function and the collision matrix, while in their analysis, some assumptions on the space and time derivatives are made [53]. On the other hand, we also note that some different analysis methods (e.g., the Chapman-Enskog analysis [4,54–56], Maxwell iteration [57–60], direct Taylor expansion or the Taylor expansion in Refs. [53,61,62], and recurrence equations [22,63–65] approaches) used to derive the macroscopic governing equations from the LB models are

developed from different physical and mathematical points of view, while it is still unclear whether they are equivalent to each other in obtaining these macroscopic governing equations (e.g., NSEs). In this paper, we will first present a unified framework for the modeling of the MRT-LB method. Then we also carry out a comparison among four different analysis methods, and, following the previous work [53], a general analysis on the MRT-LB method with some different methods is also performed. To implement the MRT-LB method, some elements are also provided.

II. MULTIPLE-RELAXATION-TIME LATTICE BOLTZMANN METHOD

The semidiscrete evolution equation of the MRT-LB method with the $DdQq$ (q discrete velocities in d -dimensional space, $d = 1 - 3$) lattice model (some commonly used lattice models can be found in Refs. [4,5]) is written as (see Appendix A for details)

$$\begin{aligned} f_j(\mathbf{x} + \mathbf{c}_j \Delta t, t + \Delta t) &= f_j(\mathbf{x}, t) - \Lambda_{jk} f_k^{\text{ne}}(\mathbf{x}, t) \\ &+ \Delta t \left[G_j(\mathbf{x}, t) + F_j(\mathbf{x}, t) + \frac{\Delta t}{2} \bar{D}_j F_j(\mathbf{x}, t) \right], \end{aligned} \quad (1)$$

where $f_j(\mathbf{x}, t)$ is the distribution function at position $\mathbf{x}$ in d -dimensional space and time t along the velocity $\mathbf{c}_j$, $\Lambda = (\Lambda_{jk})$ is a $q \times q$ invertible collision matrix, $f_j^{\text{ne}}(\mathbf{x}, t) = f_j(\mathbf{x}, t) - f_j^{\text{eq}}(\mathbf{x}, t)$ is the nonequilibrium distribution function, and $f_j^{\text{eq}}(\mathbf{x}, t)$ is the equilibrium distribution function. $F_j(\mathbf{x}, t)$ is the distribution function of a source or forcing term, and $G_j(\mathbf{x}, t)$ is the auxiliary source distribution function for removing additional terms. Δt is the time step, $\bar{D}_j = \partial_t + \gamma \mathbf{c}_j \cdot \nabla$ with $\gamma \in [0, 1]$ being a parameter to be determined later. Actually, compared to the evolution equation (A12) obtained from the discrete-velocity Boltzmann equation (see Appendix A), the introduction of parameter γ can make present MRT-LB method more general.

Generally, the evolution process based on Eq. (1) can be divided into two substeps, i.e., collision and propagation,

$$\begin{aligned} \text{Collision : } \tilde{f}_j(\mathbf{x}, t) &= f_j(\mathbf{x}, t) - \Lambda_{jk} f_k^{\text{ne}}(\mathbf{x}, t) \\ &+ \Delta t \left[G_j(\mathbf{x}, t) + F_j(\mathbf{x}, t) + \frac{\Delta t}{2} \bar{D}_j F_j(\mathbf{x}, t) \right], \end{aligned} \quad (2a)$$

$$\text{Propagation : } f_j(\mathbf{x} + \mathbf{c}_j \Delta t, t + \Delta t) = \tilde{f}_j(\mathbf{x}, t), \quad (2b)$$

where $\tilde{f}_j(\mathbf{x}, t)$ is the postcollision distribution function. It should be noted that Eq. (1) has the form of popular MRT-LB model with collision matrix Λ [12], and the commonly used LB models for the NSEs and NCDEs can be viewed as its special cases (see Sec. V for details). Here we also point out that in almost all of the MRT-LB models, the collision process in Eq. (1) is carried out in the moment space rather than the velocity (distribution function) space, and the analysis method (e.g., the Chapman-Enskog analysis) is usually more complicated and often depends on the specified lattice structure or discrete velocity set [4,12,17,66,67]. In this work, we aim to provide a unified framework for the modeling and theoretical analysis of the MRT-LB method (1) in the velocity space.

In the MRT-LB method, besides the evolution equation (1), some other key elements including the equilibrium distribution function f_j^{eq} , distribution function of the source or forcing term F_j , collision matrix Λ , and auxiliary source distribution function G_j , must also be given properly. In addition, in the implementation of the MRT-LB method, there are two popular schemes that can be used to discretize the last term on the right-hand side of Eq. (1), i.e., $\Delta t \bar{D}_j F_j(\mathbf{x}, t)/2$. Actually, if $\gamma = 0$, then the first-order explicit difference scheme $\partial_t F_j(\mathbf{x}, t) = [F_j(\mathbf{x}, t) - F_j(\mathbf{x}, t - \Delta t)]/\Delta t$ is adopted for NCDEs [17,68]. While for the case of $\gamma = 1$, we can use the first-order implicit difference scheme $(\partial_t + \mathbf{c}_j \cdot \nabla) F_j(\mathbf{x}, t) = [F_j(\mathbf{x} + \mathbf{c}_j \Delta t, t + \Delta t) - F_j(\mathbf{x}, t)]/\Delta t$, and then the linear transform considered in Refs. [17,69,70] is applied to avoid the implicitness.

III. THE ANALYSIS OF THE MRT-LB METHOD: CHAPMAN-ENSKOG ANALYSIS, MAXWELL ITERATION, DIRECT TAYLOR EXPANSION, AND RECURRENCE EQUATIONS

In the literature, there are mainly four basic analysis approaches for LB models, i.e., (1) Chapman-Enskog (CE) analysis, (2) the Maxwell iteration (MI) method, (3) the direct Taylor expansion (DTE) method, and (4) the recurrence equations (RE) method.

In the above four approaches, the CE analysis is the most widely used method in the LB community and combines the physical principle with mathematical technique (Taylor

expansion). Different from the CE analysis, however, in the MI, DTE, and RE methods, only the Taylor expansion is adopted. In what follows, we will present an introduction to these four analysis methods and also conduct a comparison among them.

A. The Chapman-Enskog analysis

In the LB method, there are two basic versions of CE expansion [2,4,17,55,71], but one can show that they are equivalent to each other. Here to derive the macroscopic equation from Eq. (1), we only consider the popular CE analysis in which the distribution functions f_j , F_j , G_j , and the time and space derivatives are expanded as [2,4,17,55,56]

$$f_j = f_j^{(0)} + \epsilon f_j^{(1)} + \epsilon^2 f_j^{(2)}, \quad (3a)$$

$$G_j = \epsilon G_j^{(1)} + \epsilon^2 G_j^{(2)}, F_j = \epsilon F_j^{(1)} + \epsilon^2 F_j^{(2)}, \quad (3b)$$

$$\partial_t = \epsilon \partial_{t_1} + \epsilon^2 \partial_{t_2}, \nabla = \epsilon \nabla_1, \quad (3c)$$

where ϵ is a small expansion parameter. In the classical kinetic theory for fluid flows, it is proportional to the Knudsen number ($\text{Kn} = \lambda/L$), and λ and L are the mean free path and characteristic length, respectively.

Applying the Taylor expansion to Eq. (1), we have

$$D_j f_j + \frac{\Delta t}{2} D_j^2 f_j + \dots = -\frac{1}{\Delta t} \Lambda_{jk} f_k^{\text{ne}} + G_j + F_j + \frac{\Delta t}{2} \bar{D}_j F_j, \quad (4)$$

where $D_j = \partial_t + \mathbf{c}_j \cdot \nabla$. Denoting $D_{1j} = \partial_{t_1} + \mathbf{c}_j \cdot \nabla_1$, $\bar{D}_{1j} = \partial_{t_1} + \gamma \mathbf{c}_j \cdot \nabla_1$, and substituting Eq. (3) into Eq. (4) yields the following equations at different orders of ϵ :

$$\epsilon^0: \Lambda_{jk} [f_k^{(0)} - f_k^{\text{eq}}] = 0, \quad (5a)$$

$$\epsilon^1: D_{1j} f_j^{(0)} = -\frac{1}{\Delta t} \Lambda_{jk} f_k^{(1)} + G_j^{(1)} + F_j^{(1)}, \quad (5b)$$

$$\epsilon^2: \partial_{t_2} f_j^{(0)} + D_{1j} f_j^{(1)} + \frac{\Delta t}{2} D_{1j}^2 f_j^{(0)} = -\frac{1}{\Delta t} \Lambda_{jk} f_k^{(2)} + G_j^{(2)} + F_j^{(2)} + \frac{\Delta t}{2} \bar{D}_{1j} F_j^{(1)}. \quad (5c)$$

Based on the fact that Λ is invertible, Eq. (5a) gives

$$f_j^{(0)} = f_j^{\text{eq}}, \quad j = 0, 1, \dots, q-1. \quad (6)$$

With the help of Eq. (5b), we can rewrite Eq. (5c) as

$$\begin{aligned} \epsilon^2: \partial_{t_2} f_j^{(0)} + D_{1j} \left(\delta_{jk} - \frac{\Lambda_{jk}}{2} \right) f_k^{(1)} + \frac{\Delta t}{2} D_{1j} \left[G_j^{(1)} + F_j^{(1)} \right] \\ = -\frac{1}{\Delta t} \Lambda_{jk} f_k^{(2)} + G_j^{(2)} + F_j^{(2)} + \frac{\Delta t}{2} \bar{D}_{1j} F_j^{(1)}. \end{aligned} \quad (7)$$

In addition, multiplying ϵ on both sides of Eq. (5b) and including some terms at the order of ϵ^2 , we have

$$D_j f_j^{\text{eq}} = -\frac{1}{\Delta t} \Lambda_{jk} f_k^{\text{ne}} + G_j + F_j + O(\epsilon^2 + \epsilon^2/\Delta t). \quad (8)$$

Through a combination of Eqs. (5b) and (7), i.e., $\epsilon^1 \times \text{Eq. (5b)} + \epsilon^2 \times \text{Eq. (7)}$, we can also obtain

$$\begin{aligned} D_j f_j^{\text{eq}} + D_j \left(\delta_{jk} - \frac{\Lambda_{jk}}{2} \right) f_k^{\text{ne}} + \frac{\Delta t}{2} D_j (G_j + F_j) \\ = -\frac{1}{\Delta t} \Lambda_{jk} f_k^{\text{ne}} + G_j + F_j + \frac{\Delta t}{2} \bar{D}_j F_j + O(\epsilon^3 + \epsilon^3/\Delta t). \end{aligned} \quad (9)$$

B. The Maxwell iteration method

To simplify the following discussion on the MI method, we first introduce the following notations:

$$\begin{aligned} \mathbf{f} = (f_0, f_1, \dots, f_{q-1})^T, \quad \mathbf{f}^{\text{eq}} = (f_0^{\text{eq}}, f_1^{\text{eq}}, \dots, f_{q-1}^{\text{eq}})^T, \\ \mathbf{f}^{\text{ne}} = \mathbf{f} - \mathbf{f}^{\text{eq}}, \end{aligned} \quad (10a)$$

$$\mathbf{G} = (G_0, G_1, \dots, G_{q-1})^T, \quad \mathbf{F} = (F_0, F_1, \dots, F_{q-1})^T, \quad (10b)$$

$$f_j(\mathbf{x} + \mathbf{c}_j \Delta t, t + \Delta t) = f_j(\mathbf{x}, t) - \Lambda_{jk} f_k^{\text{ne}}(\mathbf{x}, t) + \Delta t \left[G_j(\mathbf{x}, t) + F_j(\mathbf{x}, t) + \frac{\Delta t}{2} \bar{D}_j F_j(\mathbf{x}, t) \right], \quad (1)$$

$$\bar{\mathbf{D}} = \text{diag}(\bar{D}_0, \bar{D}_1, \dots, \bar{D}_{q-1}),$$

$$\mathbb{D} = \text{diag}(D_0, D_1, \dots, D_{q-1}), \quad (10c)$$

$$\mathbf{L} = \mathbf{L}(\Delta t \mathbb{D}) = \sum_{k \geq 1} \frac{(\Delta t \mathbb{D})^k}{k!}, \quad (10d)$$

and then with the Taylor expansion, Eq. (1) can be written as an operator equation,

$$\mathbf{L}\mathbf{f} = -\Lambda \mathbf{f}^{\text{ne}} + \Delta t \tilde{\mathbf{F}} = -\Lambda(\mathbf{f} - \mathbf{f}^{\text{eq}}) + \Delta t \tilde{\mathbf{F}}, \quad (11)$$

where $\tilde{\mathbf{F}} = \mathbf{G} + \mathbf{F} + (\Delta t/2)\bar{\mathbf{D}}\mathbf{F}$.

From Eq. (11), one can obtain

$$\mathbf{f} = \mathbf{f}^{\text{eq}} + \Delta t \Lambda^{-1} \tilde{\mathbf{F}} - \Lambda^{-1} \mathbf{L}\mathbf{f}. \quad (12)$$

Denote

$$\bar{\mathbf{L}} = \Lambda^{-1} \mathbf{L}, \quad \bar{\mathbf{G}} = \Lambda^{-1} \tilde{\mathbf{F}},$$

$$\bar{\mathbf{f}}^{\text{eq}} = \mathbf{f}^{\text{eq}} + \Delta t \Lambda^{-1} \tilde{\mathbf{F}} = \mathbf{f}^{\text{eq}} + \Delta t \bar{\mathbf{G}}, \quad (13)$$

and we can write Eq. (12) in a simple form,

$$\mathbf{f} = \bar{\mathbf{f}}^{\text{eq}} - \bar{\mathbf{L}}\mathbf{f}. \quad (14)$$

In the MI method [57–60] where the distribution function $\mathbf{f}$ is substituted into the right-hand side of Eq. (14), one can obtain the following formula from Eq. (14):

$$\mathbf{f} = \sum_{i=0}^{+\infty} (-\bar{\mathbf{L}})^i \bar{\mathbf{f}}^{\text{eq}} = \sum_{i=0}^k (-\bar{\mathbf{L}})^i \bar{\mathbf{f}}^{\text{eq}} + O(\Delta t^{k+1})$$

$$= \sum_{i=0}^k (-\bar{\mathbf{L}})^i \mathbf{f}^{\text{eq}} + \Delta t \sum_{i=0}^{k-1} (-\bar{\mathbf{L}})^i \bar{\mathbf{G}} + O(\Delta t^{k+1}), \quad (15)$$

which also leads to

$$\mathbf{f} = (\mathbf{I} - \bar{\mathbf{L}}) \mathbf{f}^{\text{eq}} + \Delta t \bar{\mathbf{G}}$$

$$= (\mathbf{I} - \bar{\mathbf{L}}) \mathbf{f}^{\text{eq}} + \Delta t \Lambda^{-1} (\mathbf{G} + \mathbf{F}) + O(\Delta t^2), \quad (16a)$$

$$\mathbf{f} = (\mathbf{I} - \bar{\mathbf{L}} + \bar{\mathbf{L}}^2) \mathbf{f}^{\text{eq}} + \Delta t (\mathbf{I} - \bar{\mathbf{L}}) \Lambda^{-1} (\mathbf{G} + \mathbf{F}) + \frac{\Delta t^2}{2} \Lambda^{-1} \bar{\mathbf{D}} \mathbf{F} + O(\Delta t^3), \quad (16b)$$

where $\bar{\mathbf{L}} = O(\Delta t)$ has been applied.

In addition, from Eq. (10) one can also derive

$$\bar{\mathbf{L}} = \Lambda^{-1} \mathbf{L} = \Lambda^{-1} \left(\Delta t \mathbb{D} + \frac{\Delta t^2}{2!} \mathbb{D}^2 + \frac{\Delta t^3}{3!} \mathbb{D}^3 + \dots \right)$$

$$= \Lambda^{-1} \left(\Delta t \mathbb{D} + \frac{\Delta t^2}{2!} \mathbb{D}^2 \right) + O(\Delta t^3)$$

$$= \Delta t \Lambda^{-1} \mathbb{D} + O(\Delta t^2), \quad (17a)$$

$$\bar{\mathbf{L}}^2 = \Delta t^2 (\Lambda^{-1} \mathbb{D})^2 + O(\Delta t^3). \quad (17b)$$

Substituting Eq. (17) into Eq. (16) yields

$$\mathbf{f}^{\text{ne}} = \mathbf{f} - \mathbf{f}^{\text{eq}} = -\Delta t \Lambda^{-1} \mathbb{D} \mathbf{f}^{\text{eq}} + \Delta t \Lambda^{-1} (\mathbf{G} + \mathbf{F}) + O(\Delta t^2), \quad (18a)$$

$$\mathbf{f}^{\text{ne}} = \mathbf{f} - \mathbf{f}^{\text{eq}} = -\Delta t \Lambda^{-1} \mathbb{D} \mathbf{f}^{\text{eq}} + \Delta t \bar{\mathbf{G}} + \Delta t^2 \Lambda^{-1} \mathbb{D} \bar{\mathbf{G}}$$

$$\times \left[\left(\mathbf{I} - \frac{\Lambda}{2} \right) \Lambda^{-1} \mathbb{D} \mathbf{f}^{\text{eq}} - \Lambda^{-1} (\mathbf{G} + \mathbf{F}) \right] + O(\Delta t^3). \quad (18b)$$

Based on Eq. (18) and using $\bar{\mathbf{G}} = \Lambda^{-1} \tilde{\mathbf{F}} = \Lambda^{-1} (\mathbf{G} + \mathbf{F}) + O(\Delta t)$, we have [72]

$$\mathbf{f}^{\text{ne}} = O(\Delta t), \quad (19a)$$

$$\mathbb{D} \mathbf{f}^{\text{eq}} = -\frac{\Lambda}{\Delta t} \mathbf{f}^{\text{ne}} + \mathbf{G} + \mathbf{F} + O(\Delta t), \quad (19b)$$

$$\mathbb{D} \mathbf{f}^{\text{eq}} - \Delta t \mathbb{D} \left[\left(\mathbf{I} - \frac{\Lambda}{2} \right) \Lambda^{-1} \mathbb{D} \mathbf{f}^{\text{eq}} - \Lambda^{-1} (\mathbf{G} + \mathbf{F}) \right] = -\frac{\Lambda}{\Delta t} \mathbf{f}^{\text{ne}} + \mathbf{G} + \mathbf{F} + \frac{\Delta t}{2} \bar{\mathbf{D}} \mathbf{F} + O(\Delta t^2). \quad (19c)$$

Inserting Eq. (19b) into Eq. (19c) gives rise to the following equation:

$$\mathbb{D} \mathbf{f}^{\text{eq}} + \mathbb{D} \left(\mathbf{I} - \frac{\Lambda}{2} \right) \mathbf{f}^{\text{ne}} + \frac{\Delta t}{2} \mathbb{D} (\mathbf{G} + \mathbf{F})$$

$$= -\frac{\Lambda}{\Delta t} \mathbf{f}^{\text{ne}} + \mathbf{G} + \mathbf{F} + \frac{\Delta t}{2} \bar{\mathbf{D}} \mathbf{F} + O(\Delta t^2). \quad (20)$$

We note that Eqs. (19b) and (20) are the same as Eqs. (8) and (9) in the CE analysis except the truncation errors neglected. Actually, when $\epsilon = O(\Delta t)$, the truncation errors of the CE analysis and MI method are of the same order in Δt .

C. The direct Taylor expansion method

From the above discussion, it can be seen clearly that Eqs. (20) and (9) have a similar form. In addition, one can also find that the aim of the MI method is to give an approximate

expression of distribution function in terms of $\bar{\mathbf{f}}^{\text{eq}}$ or $\mathbf{f}^{\text{eq}}$ [see Eq. (15)], which usually makes the MI method a little more complicated than the CE expansion. However, above analysis [see Eqs. (19) and (20)] based on the MI method also implies that it may be possible to get the same results by directly using the Taylor expansion. In the following, we present this simple approach, called the direct Taylor expansion method, which is similar to those adopted in previous works [53,61,62].

The basic idea of the DTE method is to give a series of equations on $\mathbf{f}^{\text{eq}}$ and $\mathbf{f}^{\text{ne}}$ with different orders of truncation errors from Eq. (1), which is different from that in the MI method. If we apply the Taylor expansion to Eq. (1), then one can get

$$\sum_{l=1}^N \frac{\Delta t^l}{l!} D_{jj}^l f_j + O(\Delta t^{N+1}) = -\Lambda_{jk} f_k^{\text{ne}} + \Delta t \tilde{F}_j, \quad (21)$$

Based on the relation $f_j = f_j^{\text{eq}} + f_j^{\text{ne}}$ and Eq. (21), the following equations are obtained:

$$f_j^{\text{ne}} = O(\Delta t), \quad (22a)$$

$$\sum_{l=1}^{N-1} \frac{\Delta t^l}{l!} D_j^l (f_j^{\text{eq}} + f_j^{\text{ne}}) + \frac{\Delta t^N}{N!} D_j^N f_j^{\text{eq}} = -\Lambda_{jk} f_k^{\text{ne}} + \Delta t \tilde{F}_j + O(\Delta t^{N+1}). \quad (22b)$$

Then from Eq. (22), one can derive the equations at different orders of Δt ,

$$\Delta t D_j f_j^{\text{eq}} = -\Lambda_{jk} f_k^{\text{ne}} + \Delta t (G_j + F_j) + O(\Delta t^2), \quad (23a)$$

$$\Delta t D_j (f_j^{\text{eq}} + f_j^{\text{ne}}) + \frac{\Delta t^2}{2} D_j^2 f_j^{\text{eq}} = -\Lambda_{jk} f_k^{\text{ne}} + \Delta t \left(G_j + F_j + \frac{\Delta t}{2} \bar{D}_j F_j \right) + O(\Delta t^3), \quad (23b)$$

or, equivalently,

$$D_j f_j^{\text{eq}} = -\frac{\Lambda_{jk}}{\Delta t} f_k^{\text{ne}} + G_j + F_j + O(\Delta t), \quad (24a)$$

$$D_j (f_j^{\text{eq}} + f_j^{\text{ne}}) + \frac{\Delta t}{2} D_j^2 f_j^{\text{eq}} = -\frac{\Lambda_{jk}}{\Delta t} f_k^{\text{ne}} + G_j + F_j + \frac{\Delta t}{2} \bar{D}_j F_j + O(\Delta t^2). \quad (24b)$$

According to Eq. (23a), we have

$$D_j f_j^{\text{ne}} = -\Delta t D_j \Lambda_{jk}^{-1} [D_k f_k^{\text{eq}} - (G_k + F_k)] + O(\Delta t^2), \quad (25a)$$

$$\frac{\Delta t}{2} D_j^2 f_j^{\text{eq}} = -\frac{1}{2} D_j \Lambda_{jk} f_k^{\text{ne}} + \frac{\Delta t}{2} D_j (G_j + F_j) + O(\Delta t^2). \quad (25b)$$

Substituting Eq. (25) into Eq. (24b), one can obtain the following two equivalent equations:

$$D_j f_j^{\text{eq}} - \Delta t D_j \left[\left(\delta_{jl} - \frac{\Lambda_{jl}}{2} \right) \Lambda_{lk}^{-1} D_k f_k^{\text{eq}} - \Lambda_{jk}^{-1} (G_k + F_k) \right] = -\frac{\Lambda_{jk}}{\Delta t} f_k^{\text{ne}} + G_j + F_j + \frac{\Delta t}{2} \bar{D}_j F_j + O(\Delta t^2), \quad (26)$$

$$D_j f_j^{\text{eq}} + D_j \left(\delta_{jk} - \frac{\Lambda_{jk}}{2} \right) f_k^{\text{ne}} + \frac{\Delta t}{2} D_j (G_j + F_j) = -\frac{\Lambda_{jk}}{\Delta t} f_k^{\text{ne}} + G_j + F_j + \frac{\Delta t}{2} \bar{D}_j F_j + O(\Delta t^2). \quad (27)$$

It is worth noting that Eqs. (24a) and (27) are the same as Eqs. (19b) and (20), which means that the DTE method is equivalent to the MI method at the first and second orders of time step, while the DTE method seems more straightforward and much simpler. In addition, we also point out that, different from the DTE method used in Refs. [53,62], we do not make any assumptions in above analysis.

D. The recurrence equations method

In the RE method [63–65], a kind of difference equations about f_j^{eq} and f_j^{ne} named recurrence equations is first constructed from the evolution equation of LB model, and then one can apply Taylor expansion to these recurrence equations to obtain the series equations at different orders of truncation errors. Based on this method, one can also conduct some theoretical analysis on the accuracy and stability of LB method [22].

In order to derive REs, we first rewrite Eq. (1) as

$$f_j(\mathbf{x} + \mathbf{c}_j \Delta t, t) = \tilde{f}_j(\mathbf{x}, t - \Delta t), \quad (28a)$$

$$\tilde{f}_j(\mathbf{x} - \mathbf{c}_j \Delta t, t) = f_j(\mathbf{x}, t + \Delta t). \quad (28b)$$

Let

$$\mathbf{g} = -\Lambda \mathbf{f}^{\text{ne}}, \quad \bar{\mathbf{g}} = \mathbf{g} + \Delta t \tilde{\mathbf{F}} = -\Lambda \bar{\mathbf{f}}^{\text{ne}} = -\Lambda (\mathbf{f} - \mathbf{f}^{\text{eq}}), \quad \Lambda^{-1} = \mathbf{I}/2 + \bar{\Lambda}, \quad (29)$$

and then

$$\mathbf{f}^{\text{ne}} = -\Lambda^{-1} \mathbf{g}, \quad \bar{\mathbf{f}}^{\text{ne}} = -\Lambda^{-1} \bar{\mathbf{g}}, \quad \mathbf{I} - \Lambda^{-1} = \mathbf{I}/2 - \bar{\Lambda}, \quad (30)$$

which can be used to derive the following equations:

$$\begin{aligned} f_j &= \tilde{f}_j^{\text{eq}} + \tilde{f}_j^{\text{ne}} = \tilde{f}_j^{\text{eq}} - (\Lambda^{-1})_{jk} \bar{g}_k \\ &= \tilde{f}_j^{\text{eq}} - \left(\frac{\delta_{jk}}{2} + \bar{\Lambda}_{jk} \right) \bar{g}_k, \end{aligned} \quad (31a)$$

$$\begin{aligned} \tilde{f}_j &= \tilde{f}_j^{\text{eq}} + (\delta_{jk} - \Lambda_{jk}) \tilde{f}_k^{\text{ne}} \\ &= \tilde{f}_j^{\text{eq}} + \left(\frac{\delta_{jk}}{2} - \bar{\Lambda}_{jk} \right) \bar{g}_k. \end{aligned} \quad (31b)$$

Substituting Eq. (31) into Eq. (28), one can obtain

$$\begin{aligned} &\left[\tilde{f}_j^{\text{eq}} - \left(\frac{\delta_{jk}}{2} + \bar{\Lambda}_{jk} \right) \bar{g}_k \right] (\mathbf{x} + \mathbf{c}_j \Delta t, t) \\ &= \left[\tilde{f}_j^{\text{eq}} + \left(\frac{\delta_{jk}}{2} - \bar{\Lambda}_{jk} \right) \bar{g}_k \right] (\mathbf{x}, t - \Delta t), \end{aligned} \quad (32a)$$

$$\begin{aligned} &\left[\tilde{f}_j^{\text{eq}} + \left(\frac{\delta_{jk}}{2} - \bar{\Lambda}_{jk} \right) \bar{g}_k \right] (\mathbf{x} - \mathbf{c}_j \Delta t, t) \\ &= \left[\tilde{f}_j^{\text{eq}} - \left(\frac{\delta_{jk}}{2} + \bar{\Lambda}_{jk} \right) \bar{g}_k \right] (\mathbf{x}, t + \Delta t). \end{aligned} \quad (32b)$$

Following Refs. [63,64], we first introduce the following central difference schemes for the time and space derivatives of the variable $\phi = \{\bar{f}_j^{\text{eq}}, \bar{g}_j\}$:

$$\bar{\Delta}_t \phi(\mathbf{x}, t) = [\phi(\mathbf{x}, t + \Delta t) - \phi(\mathbf{x}, t - \Delta t)]/2, \quad (33a)$$

$$\Delta_t^2 \phi(\mathbf{x}, t) = \phi(\mathbf{x}, t + \Delta t) - 2\phi(\mathbf{x}, t) + \phi(\mathbf{x}, t - \Delta t), \quad (33b)$$

$$\bar{\Delta}_j \phi(\mathbf{x}, t) = [\phi(\mathbf{x} + \mathbf{c}_j \Delta t, t) - \phi(\mathbf{x} - \mathbf{c}_j \Delta t, t)]/2, \quad (33c)$$

$$\Delta_j^2 \phi(\mathbf{x}, t) = \phi(\mathbf{x} + \mathbf{c}_j \Delta t, t) - 2\phi(\mathbf{x}, t) + \phi(\mathbf{x} - \mathbf{c}_j \Delta t, t), \quad (33d)$$

and then we can obtain

$$\bar{\Delta}_j \phi(\mathbf{x}, t) = -\bar{\Delta}_{\bar{j}} \phi(\mathbf{x}, t), \quad \Delta_j^2 \phi(\mathbf{x}, t) = \Delta_{\bar{j}}^2 \phi(\mathbf{x}, t), \quad (34)$$

where $\bar{j}$ is the opposite direction of j , $\mathbf{c}_j = -\mathbf{c}_{\bar{j}}$.

The sum and the difference of Eqs. (32a) and (32b) become

$$[(\Delta_j^2 - \Delta_t^2)(\bar{f}_j^{\text{eq}} - \bar{\Lambda}_{jk} \bar{g}_k) - (\bar{\Delta}_j - \bar{\Delta}_t) \bar{g}_j](\mathbf{x}, t) = 0, \quad (35a)$$

$$[(\bar{\Delta}_j + \bar{\Delta}_t)(\bar{f}_j^{\text{eq}} - \bar{\Lambda}_{jk} \bar{g}_k) - \frac{1}{4}(\Delta_j^2 + \Delta_t^2) \bar{g}_j](\mathbf{x}, t) = \bar{g}_j(\mathbf{x}, t). \quad (35b)$$

Based on Eqs. (34) and (35) where j is replaced by $\bar{j}$, one can also obtain the following equations:

$$[(\Delta_j^2 - \Delta_t^2)(\bar{f}_{\bar{j}}^{\text{eq}} - \bar{\Lambda}_{\bar{j}k} \bar{g}_k) + (\bar{\Delta}_j + \bar{\Delta}_t) \bar{g}_{\bar{j}}](\mathbf{x}, t) = 0, \quad (36a)$$

$$[(-\bar{\Delta}_j + \bar{\Delta}_t)(\bar{f}_{\bar{j}}^{\text{eq}} - \bar{\Lambda}_{\bar{j}k} \bar{g}_k) - \frac{1}{4}(\Delta_j^2 + \Delta_t^2) \bar{g}_{\bar{j}}](\mathbf{x}, t) = \bar{g}_{\bar{j}}(\mathbf{x}, t). \quad (36b)$$

The sum and the difference of Eqs. (35a) and (36a) give rise to [63,64]

$$[(\Delta_j^2 - \Delta_t^2)(\bar{f}_j^{\text{eq},+} - \bar{\Lambda}_{jk}^+ \bar{g}_k) - \bar{\Delta}_j \bar{g}_j^- + \bar{\Delta}_t \bar{g}_j^+](\mathbf{x}, t) = 0, \quad (37a)$$

$$[(\Delta_j^2 - \Delta_t^2)(\bar{f}_j^{\text{eq},-} - \bar{\Lambda}_{jk}^- \bar{g}_k) - \bar{\Delta}_j \bar{g}_j^+ + \bar{\Delta}_t \bar{g}_j^-](\mathbf{x}, t) = 0. \quad (37b)$$

Similarly, from Eqs. (35b) and (36b), one can also obtain

$$[\bar{\Delta}_j(\bar{f}_j^{\text{eq},-} - \bar{\Lambda}_{jk}^- \bar{g}_k) + \bar{\Delta}_t(\bar{f}_j^{\text{eq},+} - \bar{\Lambda}_{jk}^+ \bar{g}_k) - \frac{1}{4}(\Delta_j^2 + \Delta_t^2) \bar{g}_j^+](\mathbf{x}, t) = \bar{g}_j^+(\mathbf{x}, t), \quad (38a)$$

$$[\bar{\Delta}_j(\bar{f}_j^{\text{eq},+} - \bar{\Lambda}_{jk}^+ \bar{g}_k) + \bar{\Delta}_t(\bar{f}_j^{\text{eq},-} - \bar{\Lambda}_{jk}^- \bar{g}_k) - \frac{1}{4}(\Delta_j^2 + \Delta_t^2) \bar{g}_j^-](\mathbf{x}, t) = \bar{g}_j^-(\mathbf{x}, t), \quad (38b)$$

where $\phi_j^+ = (\phi_j + \phi_{\bar{j}})/2$ and $\phi_j^- = (\phi_j - \phi_{\bar{j}})/2$ are the symmetric and antisymmetric parts of $\phi_j = \{\bar{f}_j^{\text{eq}}, \bar{g}_j, \bar{\Lambda}_{jk}\}$ [56]. If the terms related to time difference are dropped from Eq. (35), then one can immediately derive the steady-state REs,

$$[\Delta_j^2(\bar{f}_j^{\text{eq}} - \bar{\Lambda}_{jk} \bar{g}_k) - \bar{\Delta}_j \bar{g}_j](\mathbf{x}, t) = 0, \quad (39a)$$

$$[\bar{\Delta}_j(\bar{f}_j^{\text{eq}} - \bar{\Lambda}_{jk} \bar{g}_k) - \frac{1}{4} \Delta_j^2 \bar{g}_j](\mathbf{x}, t) = \bar{g}_j(\mathbf{x}, t). \quad (39b)$$

Similarly, we can also obtain the steady-state REs corresponding to Eqs. (37) and (38). It should also be noted that, unlike previous analysis methods, the REs are exact linear combinations of the evolution equation and can be formulated directly for the steady-state problems [73–75]. Besides, the REs can also be used to obtain the equivalent difference equations and exact solutions of the LB method and analyze the boundary effects.

In addition, we point out that Eqs. (35), (36), (37), and (38) are the so-called REs obtained from Eq. (1). Through using Taylor expansion to the REs, we can also determine a series of equations on f_j^{eq} and g_j (or f_j^{ne}) with different orders of time step. Actually, applying the Taylor expansion to Eq. (35b) gives

$$D_j(\bar{f}_j^{\text{eq}} - \bar{\Lambda}_{jk} \bar{g}_k) = \frac{1}{\Delta t} \bar{g}_j + O(\Delta t^2), \quad (40)$$

Based on the relation $\bar{g}_j = O(\Delta t)$ [see Eq. (29)], we have

$$D_j \bar{f}_j^{\text{eq}} = \frac{1}{\Delta t} \bar{g}_j + O(\Delta t). \quad (41)$$

Then we can also rewrite Eqs. (41) and (40) as

$$D_j f_j^{\text{eq}} = \frac{1}{\Delta t} g_j + G_j + F_j + O(\Delta t), \quad (42a)$$

$$D_j(f_j^{\text{eq}} - \bar{\Lambda}_{jk} g_k) + \frac{\Delta t}{2} D_j \bar{F}_j = \frac{1}{\Delta t} g_j + \bar{F}_j + O(\Delta t^2), \quad (42b)$$

where Eqs. (13), (29), and (30) have been used. Keeping in mind that $\bar{F}_j = G_j + F_j + (\Delta t/2) \bar{D}_j F_j$, the following equation can be obtained from Eq. (42b):

$$D_j f_j^{\text{eq}} - D_j \bar{\Lambda}_{jk} g_k + \frac{\Delta t}{2} D_j (G_j + F_j) = \frac{1}{\Delta t} g_j + G_j + F_j + \frac{\Delta t}{2} \bar{D}_j F_j + O(\Delta t^2). \quad (43)$$

It should be noted that Eqs. (42a) and (43) are the same as Eqs. (24a) and (27) without considering the truncation errors, which also means that the MI, DTE, and RE methods are equivalent to each other at the first and second orders of time step.

Finally, we also give some general discussion on the CE analysis and the MI, DTE, and RE methods. On the one hand, it is well known that the CE analysis is a basic method in the kinetic theory and LB community, while in this method, we need to introduce one another parameter ϵ . For fluid flow problems, the parameter ϵ is proportional to the Knudsen number, but for the problems described by NCDEs, it may have no any physical meaning. On the other hand, from the mathematical point of view, the other three (MI, DTE, and

RE) methods only adopt the Taylor expansion and should be equivalent to each other, while, among them, the DTE method seems much simpler.

IV. DERIVATION OF THE MACROSCOPIC EQUATIONS

In this section, we will adopt the above analysis methods to derive the macroscopic equations including the NCDE and classical NSEs and mainly focus on the CE analysis and some of other methods. This is because from the discussion in the previous section, we can see that the CE method has incorporated some physical principles, and compared to some other mathematical methods based on the Taylor expansion, it also presents some different results at the higher-order equations. On the other hand, it is clear that the MI, DTE, and RE methods are equivalent to each other since they can give the same equations at the first and second orders of time step, and thus, for brevity, only some of them are considered.

A. Derivation of NCDE

The d -dimensional NCDE with a source term can be expressed as [17]

$$\partial_t \phi + \nabla \cdot \mathbf{B} = \nabla \cdot [\mathbf{K} \cdot (\nabla \cdot \mathbf{D})] + S, \quad (44)$$

where ϕ is a unknown scalar function of position $\mathbf{x}$ and time t , S is a scalar source term. $\mathbf{B} = (\mathbf{B}_\alpha)$ is a vector function, $\mathbf{K} = (\mathbf{K}_{\alpha\beta})$ and $\mathbf{D} = (\mathbf{D}_{\alpha\beta})$ are symmetric tensors (matrices), and they can be functions of ϕ , $\mathbf{x}$, and t . We note that Eq. (44) can be considered a general form of CDEs, and many different kinds of CDEs considered in some previous work [19,68,76] are its special cases. When $\mathbf{K}_{\alpha\beta} = \kappa \delta_{\alpha\beta}$ and $\mathbf{D}_{\alpha\beta} = h \delta_{\alpha\beta}$ with κ and h being two scalar functions, Eq. (44) becomes an isotropic CDE.

To recover Eq. (44) from the MRT-LB method (1), we only need to give some appropriate constraints on the collision matrix $\mathbf{A}$, the moments of $f_j^{\text{eq}}(\mathbf{x}, t)$, $G_j(\mathbf{x}, t)$, and $F_j(\mathbf{x}, t)$. Based on these constraints, one can determine the expressions of $\mathbf{A}$ and these distribution functions. Actually, to derive Eq. (44), only the zeroth- to second-order moments of f_j^{eq} , the zeroth- and first-order moments of G_j and F_j , and the constraints on $\mathbf{A}$ corresponding to these moments are needed. In the MRT-LB method, the unknown conserved scalar ϕ can be calculated by $\phi = \sum_j f_j$, and the distribution functions f_j , f_j^{eq} , G_j and F_j should satisfy the following relations:

$$\sum_j f_j = \sum_j f_j^{\text{eq}} = \phi, \quad \sum_j \mathbf{c}_j f_j^{\text{eq}} = \mathbf{B},$$

$$\sum_j \mathbf{c}_j \mathbf{c}_j f_j^{\text{eq}} = \beta c_s^2 \mathbf{D} + \mathbf{C}, \quad (45a)$$

$$\sum_j G_j = 0, \quad \sum_j \mathbf{c}_j G_j = \mathbf{M}_{1G}, \quad (45b)$$

$$\sum_j F_j = S, \quad \sum_j \mathbf{c}_j F_j = \mathbf{M}_{1F}, \quad (45c)$$

where c_s is a model parameter related to lattice speed $c = \Delta x / \Delta t$ with Δx being the lattice spacing. β is a parameter that can be used to adjust the relaxation matrix [see Eq. (53)],

$\mathbf{C}$ is an auxiliary moment to be determined later, and $\mathbf{M}_{1G}$ and $\mathbf{M}_{1F}$ are the first-order moments of G_j and F_j .

In addition, for the collision matrix $\mathbf{A}$, the following requirements are needed:

$$\sum_j \mathbf{e}_j \mathbf{A}_{jk} = s_0 \mathbf{e}_k, \quad \sum_j \mathbf{c}_j \mathbf{A}_{jk} = \mathbf{S}_{10} \mathbf{e}_k + \mathbf{S}_1 \mathbf{c}_k, \quad \forall k, \quad (46)$$

where $\mathbf{e} = (1, 1, \dots, 1) \in R^d$, $\mathbf{S}_{10}$ is a $d \times 1$ matrix, and $\mathbf{S}_1$ is an invertible $d \times d$ relaxation matrix corresponding to the diffusion matrix $\mathbf{K}$. Based on Eqs. (45a) and (46), one can find that the relaxation parameter s_0 does not appear in the recovered macroscopic equation, and its value can be chosen arbitrarily.

1. Derivation of the NCDE through CE analysis

In this part, we will present some details on how to derive NCDE (44) from the MRT-LB method (1). In the CE analysis, we can first express the source term as $S = \epsilon S^{(1)} + \epsilon^2 S^{(2)}$, and then from Eqs. (3), (6), and (45) we have

$$\sum_j f_j^{(k)} = 0, \quad (47a)$$

$$\sum_j G_j^{(k)} = 0, \quad \sum_j \mathbf{c}_j G_j^{(k)} = \mathbf{M}_{1G}^{(k)}, \quad (47b)$$

$$\sum_j F_j^{(k)} = S^{(k)}, \quad \sum_j \mathbf{c}_j F_j^{(k)} = \mathbf{M}_{1F}^{(k)}, \quad (47c)$$

where $k \geq 1$. Summing Eqs. (5b) and (7) over j and using Eqs. (45), (46), and (47), one can obtain

$$\partial_t \phi + \nabla_1 \cdot \mathbf{B} = S^{(1)}, \quad (48a)$$

$$\partial_t \phi + \nabla_1 \cdot \left[(\mathbf{I} - \mathbf{S}_1/2) \sum_k \mathbf{c}_k f_k^{(1)} \right] = S^{(2)} + \frac{\Delta t}{2} \nabla_1 \cdot [(\gamma - 1) \mathbf{M}_{1F}^{(1)} - \mathbf{M}_{1G}^{(1)}]. \quad (48b)$$

With the aid of Eqs. (5b), (45), (46), and (47), we get

$$\begin{aligned} & \sum_k \mathbf{c}_k f_k^{(1)} \\ &= -\Delta t \sum_k \mathbf{c}_k \mathbf{A}_{kl}^{-1} [D_{1l} f_l^{(0)} - G_l^{(1)} - F_l^{(1)}] \\ &= -\Delta t \mathbf{S}_1^{-1} \sum_l \mathbf{c}_l [D_{1l} f_l^{(0)} - G_l^{(1)} - F_l^{(1)}] \\ &= -\Delta t \mathbf{S}_1^{-1} [\partial_t \mathbf{B} + \nabla_1 \cdot (\beta c_s^2 \mathbf{D} + \mathbf{C}) - \mathbf{M}_{1G}^{(1)} - \mathbf{M}_{1F}^{(1)}]. \end{aligned} \quad (49)$$

Substituting above equation into Eq. (48b), we have

$$\partial_t \phi = \nabla_1 \cdot [\Delta t \beta c_s^2 (\mathbf{S}_1^{-1} - \mathbf{I}/2) \nabla_1 \cdot \mathbf{D}] + S^{(2)} + \Delta t \nabla_1 \cdot RH, \quad (50)$$

where

$$\begin{aligned} RH &= (\mathbf{S}_1^{-1} - \mathbf{I}/2) (\partial_t \mathbf{B} + \nabla_1 \cdot \mathbf{C}) \\ &\quad - (\mathbf{S}_1^{-1} - \gamma \mathbf{I}/2) \mathbf{M}_{1F}^{(1)} - \mathbf{S}_1^{-1} \mathbf{M}_{1G}^{(1)}. \end{aligned} \quad (51)$$

If we take $RH = 0$, then Eq. (50) would reduce to

$$\partial_t \phi = \nabla_1 \cdot [\mathbf{K} \cdot (\nabla_1 \cdot \mathbf{D})] + S^{(2)}, \quad (52)$$

with

$$\mathbf{K} = \Delta t \beta c_s^2 (\mathbf{S}_1^{-1} - \mathbf{I}/2). \quad (53)$$

Combining Eqs. (48a) with (52), we obtain the macroscopic NCDE (44).

Here we point out that in the expression of diffusion matrix $\mathbf{K}$ [see Eq. (53)], the part $(-\Delta t \beta c_s^2 \mathbf{I}/2)$ is the numerical diffusion caused by the discrete effect. Actually, when we use the CE analysis to obtain the NCDE from the discrete-velocity Boltzmann equation (see Appendix B for details), the term $(-\Delta t \beta c_s^2 \mathbf{I}/2)$ is not included in the matrix $\mathbf{K}$. In addition, from above CE analysis, we can also see that the functions $\mathbf{C}$, $\mathbf{M}_{1G}$, and $\mathbf{M}_{1F}$ must be chosen properly to make $RH = 0$, and in this case, the NCDE (44) can be recovered correctly without additional assumptions. In the following, we present some special cases.

Case 1: $\mathbf{C} = 0$, $\mathbf{M}_{1F} = 0$. Under the condition of $RH = 0$, we have

$$RH = [(\mathbf{S}_1^{-1} - \mathbf{I}/2) \partial_t \mathbf{B} - \mathbf{S}_1^{-1} \mathbf{M}_{1G}^{(1)}] = 0, \quad (54)$$

which leads to the following $\mathbf{M}_{1G}$:

$$\mathbf{M}_{1G} = (\mathbf{I} - \mathbf{S}_1/2) \partial_t \mathbf{B}. \quad (55)$$

Case 2: $\mathbf{B} = \mathbf{B}(\phi)$, $\mathbf{M}_{1F} = 0$. If $\mathbf{B}$ is a differentiable function of ϕ , then $\mathbf{C}$ and $\mathbf{M}_{1G}$ should satisfy the following equation:

$$RH = [(\mathbf{S}_1^{-1} - \mathbf{I}/2)(\partial_t \mathbf{B} + \nabla_1 \cdot \mathbf{C}) - \mathbf{S}_1^{-1} \mathbf{M}_{1G}^{(1)}] = 0, \quad (56)$$

which leads to

$$\mathbf{M}_{1G}^{(1)} = (\mathbf{I} - \mathbf{S}_1/2)(\partial_t \mathbf{B} + \nabla_1 \cdot \mathbf{C}). \quad (57)$$

Furthermore, if we take $\mathbf{C}$ as

$$\mathbf{C}_{\alpha\beta} = \int \mathbf{B}'_{\alpha} \mathbf{B}'_{\beta} d\phi, \quad (58)$$

which satisfies

$$\mathbf{C}'_{\alpha\beta} = \mathbf{B}'_{\alpha} \mathbf{B}'_{\beta}, \quad (59)$$

then it follows from Eq. (48a) that

$$\partial_t \mathbf{B} + \nabla_1 \cdot \mathbf{C} = \mathbf{B}'(\partial_t \phi + \nabla_1 \cdot \mathbf{B}) = \mathbf{B}' S^{(1)}. \quad (60)$$

Substituting the above equation into Eq. (56), one can determine $\mathbf{M}_{1G}$ by

$$\mathbf{M}_{1G} = (\mathbf{I} - \mathbf{S}_1/2) \mathbf{B}' S. \quad (61)$$

2. Derivation of NCDE through the MI method

Based on the similarity between CE analysis and the MI method, we can also recover NCDE (44) through the MI method. To do this, we first introduce the following notations:

$$\mathbf{E} = (\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_{q-1}), \quad (62a)$$

$$\langle \mathbf{E} \mathbf{E} \rangle = (\mathbf{c}_0 \mathbf{c}_0, \mathbf{c}_1 \mathbf{c}_1, \dots, \mathbf{c}_{q-1} \mathbf{c}_{q-1}), \quad (62b)$$

and then Eqs. (45) and (46) can be written in the matrix forms,

$$\phi = \mathbf{e} \cdot \mathbf{f} = \mathbf{e} \cdot \mathbf{f}^{\text{eq}}, \quad \mathbf{E} \mathbf{f}^{\text{eq}} = \mathbf{B}, \quad \langle \mathbf{E} \mathbf{E} \rangle \mathbf{f}^{\text{eq}} = \beta c_s^2 \mathbf{D} + \mathbf{C}, \quad (63a)$$

$$\mathbf{e} \cdot \mathbf{G} = 0, \quad \mathbf{E} \mathbf{G} = \mathbf{M}_{1G}, \quad \mathbf{e} \cdot \mathbf{F} = S, \quad \mathbf{E} \mathbf{F} = \mathbf{M}_{1F} = \mathbf{0}, \quad (63b)$$

$$\mathbf{e} \mathbf{A} = s_0 \mathbf{e}, \quad \mathbf{E} \mathbf{A} = \mathbf{S}_{10} \mathbf{e} + \mathbf{S}_1 \mathbf{E}. \quad (63c)$$

According to the above equations, we have

$$\mathbf{e} \cdot \mathbf{f}^{\text{ne}} = 0, \quad (64a)$$

$$\mathbf{e} \cdot \mathbb{D} \mathbf{f}^{\text{eq}} = \partial_t \mathbf{e} \cdot \mathbf{f}^{\text{eq}} + \nabla \cdot \mathbf{E} \mathbf{f}^{\text{eq}} = \partial_t \phi + \nabla \cdot \mathbf{B}, \quad (64b)$$

$$\begin{aligned} \mathbf{e} \cdot \mathbb{D}(\mathbf{G} + \mathbf{F}) &= \partial_t \mathbf{e} \cdot (\mathbf{G} + \mathbf{F}) + \nabla \cdot \mathbf{E}(\mathbf{G} + \mathbf{F}) \\ &= \partial_t S + \nabla \cdot \mathbf{M}_{1G}, \end{aligned} \quad (64c)$$

$$\mathbf{e} \cdot \bar{\mathbf{D}} \mathbf{F} = \partial_t \mathbf{e} \cdot \mathbf{F} + \gamma \nabla \cdot \mathbf{E} \mathbf{F} = \partial_t S, \quad (64d)$$

$$\begin{aligned} \mathbf{E} \mathbb{D} \mathbf{f}^{\text{eq}} &= \partial_t \mathbf{E} \mathbf{f}^{\text{eq}} + \nabla \cdot \langle \mathbf{E} \mathbf{E} \rangle \mathbf{f}^{\text{eq}} \\ &= \partial_t \mathbf{B} + \nabla \cdot (\beta c_s^2 \mathbf{D} + \mathbf{C}), \end{aligned} \quad (64e)$$

$$\begin{aligned} \mathbf{e} \cdot \mathbb{D}(\mathbf{I} - \mathbf{A}/2) \mathbf{f}^{\text{ne}} &= \partial_t \mathbf{e} \cdot (\mathbf{I} - \mathbf{A}/2) \mathbf{f}^{\text{ne}} + \nabla \cdot \mathbf{E}(\mathbf{I} - \mathbf{A}/2) \mathbf{f}^{\text{ne}} \\ &= \partial_t (1 - s_0/2) \mathbf{e} \cdot \mathbf{f}^{\text{ne}} + \nabla \cdot (\mathbf{I} - \mathbf{S}_1/2) \mathbf{E} \mathbf{f}^{\text{ne}} \\ &= \nabla \cdot (\mathbf{I} - \mathbf{S}_1/2) \mathbf{E} \mathbf{f}^{\text{ne}}. \end{aligned} \quad (64f)$$

Multiplying $\mathbf{e}$ on both sides of Eqs. (19b) and (20), and using Eq. (64), we have

$$\begin{aligned} \mathbf{e} \cdot \mathbb{D} \mathbf{f}^{\text{eq}} &= -\frac{1}{\Delta t} \mathbf{e} \cdot \mathbf{A} \mathbf{f}^{\text{ne}} + \mathbf{e} \cdot \mathbf{G} + \mathbf{e} \cdot \mathbf{F} + O(\Delta t) \\ &= -\frac{s_0}{\Delta t} \mathbf{e} \cdot \mathbf{f}^{\text{ne}} + S + O(\Delta t), \end{aligned} \quad (65a)$$

$$\begin{aligned} \mathbf{e} \cdot \mathbb{D} \mathbf{f}^{\text{eq}} + \mathbf{e} \cdot \mathbb{D}(\mathbf{I} - \mathbf{A}/2) \mathbf{f}^{\text{ne}} + \frac{\Delta t}{2} \mathbf{e} \cdot \mathbb{D}(\mathbf{G} + \mathbf{F}) \\ &= -\frac{1}{\Delta t} \mathbf{e} \mathbf{A} \mathbf{f}^{\text{ne}} + \mathbf{e} \cdot \mathbf{G} + \mathbf{e} \cdot \mathbf{F} + \frac{\Delta t}{2} \mathbf{e} \cdot \bar{\mathbf{D}} \mathbf{F} + O(\Delta t^2) \\ &= -\frac{s_0}{\Delta t} \mathbf{e} \cdot \mathbf{f}^{\text{ne}} + S + \frac{\Delta t}{2} \mathbf{e} \cdot \bar{\mathbf{D}} \mathbf{F} + O(\Delta t^2). \end{aligned} \quad (65b)$$

Based on Eqs. (63) and (64), Eq. (65) can be written as

$$\partial_t \phi + \nabla \cdot \mathbf{B} = S + O(\Delta t), \quad (66a)$$

$$\begin{aligned} \partial_t \phi + \nabla \cdot \mathbf{B} + \nabla \cdot (\mathbf{I} - \mathbf{S}_1/2) \mathbf{E} \mathbf{f}^{\text{ne}} + \frac{\Delta t}{2} \nabla \cdot \mathbf{M}_{1G} \\ &= S + O(\Delta t^2). \end{aligned} \quad (66b)$$

On the other hand, Eq. (19b) gives

$$\begin{aligned} \mathbf{E} \mathbf{f}^{\text{ne}} &= -\Delta t \mathbf{E} \mathbf{A}^{-1} (\mathbb{D} \mathbf{f}^{\text{eq}} - \mathbf{G} - \mathbf{F}) + O(\Delta t^2) \\ &= -\Delta t \mathbf{S}_1^{-1} \mathbf{E} (\mathbb{D} \mathbf{f}^{\text{eq}} - \mathbf{G} - \mathbf{F}) + O(\Delta t^2) \\ &= -\Delta t \mathbf{S}_1^{-1} [\partial_t \mathbf{B} + \nabla \cdot (\beta c_s^2 \mathbf{D} + \mathbf{C}) - \mathbf{M}_{1G}] + O(\Delta t^2). \end{aligned} \quad (67)$$

Substituting Eq. (67) into Eq. (66b), one can obtain

$$\begin{aligned} \partial_t \phi + \nabla \cdot \mathbf{B} &= \nabla \cdot [\mathbf{K} \cdot (\nabla \cdot \mathbf{D})] \\ &\quad + S + \Delta t \nabla \cdot RH_1 + O(\Delta t^2), \end{aligned} \quad (68)$$

where the diffusion matrix $\mathbf{K}$ is the same as that in Eq. (53), and RH_1 is defined by

$$RH_1 = (\mathbf{S}_1^{-1} - \mathbf{I}/2)(\partial_t \mathbf{B} + \nabla \cdot \mathbf{C}) - \mathbf{S}_1^{-1} \mathbf{M}_{1G}. \quad (69)$$

Under the requirement of $RH_1 = 0$, we can derive the following condition:

$$\mathbf{M}_{1G} = (\mathbf{I} - \mathbf{S}_1/2)(\partial_t \mathbf{B} + \nabla \cdot \mathbf{C}), \quad (70)$$

which is also consistent with Eq. (57). Under the condition of Eq. (70), we can correctly recover the NCDE (44) at the order of Δt^2 ,

$$\partial_t \phi + \nabla \cdot \mathbf{B} = \nabla \cdot [\mathbf{K} \cdot (\nabla \cdot \mathbf{D})] + S + O(\Delta t^2). \quad (71)$$

We now present a special discussion on how to calculate the diffusion flux $[-\mathbf{K} \cdot (\nabla \cdot \mathbf{D})]$, which is similar to the previous work [77,78]. Actually, from Eqs. (67) and (70) we have

$$\begin{aligned} \nabla \cdot (\beta c_s^2 \mathbf{D}) &= -\frac{\mathbf{S}_1}{\Delta t} \mathbf{E} \mathbf{f}^{\text{ne}} + [\mathbf{M}_{1G} - (\partial_t \mathbf{B} + \nabla \cdot \mathbf{C})] + O(\Delta t) \\ &= -\frac{\mathbf{S}_1}{\Delta t} \mathbf{E} \mathbf{f}^{\text{ne}} - \frac{\mathbf{S}_1}{2} (\partial_t \mathbf{B} + \nabla \cdot \mathbf{C}) + O(\Delta t) \\ &= \begin{cases} -\mathbf{S}_1 (\frac{1}{\Delta t} \mathbf{E} \mathbf{f}^{\text{ne}} + \frac{1}{2} \mathbf{B}' S) + O(\Delta t), \\ \quad \text{if } \mathbf{B} = \mathbf{B}(\phi) \text{ and } \mathbf{C} = \int \mathbf{B}' \mathbf{B}' d\phi, \\ -\mathbf{S}_1 (\frac{1}{\Delta t} \mathbf{E} \mathbf{f}^{\text{ne}} + \frac{1}{2} \partial_t \mathbf{B}) + O(\Delta t), \quad \text{if } \mathbf{C} = \mathbf{0}. \end{cases} \end{aligned} \quad (72)$$

From Eq. (72) we can obtain a local scheme for the diffusion flux with a second-order accuracy in time,

$$-\mathbf{K} \cdot (\nabla \cdot \mathbf{D}) = \begin{cases} -(\mathbf{S}_1/2 - \mathbf{I})(\mathbf{E} \mathbf{f}^{\text{ne}} + \Delta t \mathbf{B}' S/2), \\ \quad \text{if } \mathbf{B} = \mathbf{B}(\phi) \text{ and } \mathbf{C} = \int \mathbf{B}' \mathbf{B}' d\phi, \\ -(\mathbf{S}_1/2 - \mathbf{I})(\mathbf{E} \mathbf{f}^{\text{ne}} + \Delta t \partial_t \mathbf{B}/2), \quad \text{if } \mathbf{C} = \mathbf{0}, \end{cases} \quad (73)$$

where Eq. (53) has been used. Additionally, if $\mathbf{D}$ is only a function of ϕ , we can also obtain a local scheme for $\nabla \phi$ from Eq. (72) or (73).

Remark: We noted that following the idea in Ref. [68], Zhang *et al.* also developed a MRT model for a general isotropic CDE in a recent work [60] and found that under the diffusive scaling ($\Delta t = \eta \Delta x^2$, η is a parameter) [67], the general isotropic CDE can be recovered correctly from the MRT model. However, it should be noted that the diffusive scaling is an assumption and also brings an additional constrain on the time step and lattice spacing. Besides, we also point out that under the diffusive scaling, the present model and the one in Ref. [17] would reduce to the MRT model in Ref. [60]. Actually, under the diffusive scaling, the last term on the right-hand side of Eq. (1) can be neglected since it is of fourth order of Δx , and also the terms $\partial_t \phi$ and $\partial_t \mathbf{B}$ would be absent in Eqs. (66a) and (67). In this case, the auxiliary moment $\mathbf{C}$ and first-order moment $\mathbf{M}_{1G}$ are not needed and can be set to be zero, which also means that the auxiliary distribution function G_i can be ignored in the evolution equation (1). Thus, under the diffusive scaling, the evolution equation (1) can reduce to that in Ref. [60]. In addition, we would also like to point out the numerical diffusion in the LB method for NCDEs [19] is also commonly skipped by the asymptotic analysis [60,67].

3. The equilibrium, auxiliary, and source distribution functions of the MRT-LB method for NCDE

As discussed above, it can be found that to recover the NCDE (44) correctly, some proper requirements on the equilibrium, auxiliary, and source distribution functions should be satisfied. Based on Eq. (45) for a general DdQq lattice model, one can determine the expressions of f_j^{eq} , G_j , and F_j . For

simplicity, we only consider the following commonly used forms:

$$f_j^{\text{eq}} = \omega_j \left[\phi + \frac{\mathbf{c}_j \cdot \mathbf{B}}{c_s^2} + \frac{(\beta c_s^2 \mathbf{D} + \mathbf{C} - c_s^2 \phi \mathbf{I}) : (\mathbf{c}_j \mathbf{c}_j - c_s^2 \mathbf{I})}{2c_s^4} \right], \quad (74a)$$

$$G_j = \omega_j \frac{\mathbf{c}_j \cdot \mathbf{M}_{1G}}{c_s^2}, \quad (74b)$$

$$F_j = \omega_j S, \quad (74c)$$

where $\mathbf{M}_{1G}$ is given by Eq. (70).

We note that when $\mathbf{D} = \phi \mathbf{I}$, the NCDE (44) would become

$$\partial_t \phi + \nabla \cdot \mathbf{B} = \nabla \cdot (\mathbf{K} \cdot \nabla \phi) + S. \quad (75)$$

If we further take $\beta = 1$ and $\mathbf{C} = \mathbf{0}$, then the equilibrium and auxiliary distribution functions f_j^{eq} and G_j defined by Eqs. (74a) and (74b) can be simplified as

$$f_j^{\text{eq}} = \omega_j \left(\phi + \frac{\mathbf{c}_j \cdot \mathbf{B}}{c_s^2} \right), \quad (76a)$$

$$G_j = \omega_j \frac{\mathbf{c}_j \cdot (\mathbf{I} - \mathbf{S}_1/2) \partial_t \mathbf{B}}{c_s^2}, \quad (76b)$$

where the term $\partial_t \mathbf{B}$ can be computed by the first-order explicit difference scheme, i.e., $\partial_t \mathbf{B} = [\mathbf{B}(\mathbf{x}, t) - \mathbf{B}(\mathbf{x}, t - \Delta t)]/\Delta t$. We note that although a first-order scheme is used for $\partial_t \mathbf{B}$, the present LB model still has a second-order accuracy in time since there is a time step Δt in front of G_i , as seen from Eq. (1). In this case, the simple DdQ2d or DdQ(2d + 1) lattice model can be adopted.

In addition, we also point out that at the diffusive scaling, Eq. (74) can be simplified by

$$f_j^{\text{eq}} = \omega_j \left[\phi + \frac{\mathbf{c}_j \cdot \mathbf{B}}{c_s^2} + \frac{(\beta \mathbf{D} - \phi \mathbf{I}) : (\mathbf{c}_j \mathbf{c}_j - c_s^2 \mathbf{I})}{2c_s^2} \right], \quad (77a)$$

$$G_j = 0, \quad (77b)$$

$$F_j = \omega_j S. \quad (77c)$$

When $\beta = 1$ and $\mathbf{D} = D(\phi) \mathbf{I}$, the equilibrium distribution function (77a) would reduce to the one in Ref. [60]. For a more general form of equilibrium distribution function (77a), we refer the reader to the work of Ginzburg [19,22,64].

B. Derivation of NSEs

We now consider the following d -dimensional NSEs with the source and forcing terms:

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = \bar{S}, \quad (78a)$$

$$\partial_t (\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p + \nabla \cdot \sigma + \bar{\mathbf{F}}, \quad (78b)$$

where the shear stress σ is defined by

$$\begin{aligned} \sigma &= \mu [\nabla \mathbf{u} + (\nabla \mathbf{u})^T] + \lambda (\nabla \cdot \mathbf{u}) \mathbf{I} \\ &= \mu \left[\nabla \mathbf{u} + (\nabla \mathbf{u})^T - \frac{2}{d} (\nabla \cdot \mathbf{u}) \mathbf{I} \right] + \mu_b (\nabla \cdot \mathbf{u}) \mathbf{I}, \end{aligned} \quad (79)$$

where μ is the dynamic viscosity and $\lambda = \mu_b - 2\mu/d$ with μ_b being the bulk viscosity [79,80].

Similarly to the derivation of NCDE, to recover NSEs from MRT-LB method (1) we also need to give some constraints on Λ , f_j , f_j^{eq} , G_j , and F_j . In addition, compared to Eqs. (45),

(46), and (47), there is another requirement on the higher-order moments of the distribution functions for NSEs. Here the following conditions should be satisfied:

$$\rho = \sum_j f_j = \sum_j f_j^{\text{eq}}, \quad \rho \mathbf{u} = \sum_j \mathbf{c}_j f_j = \sum_j \mathbf{c}_j f_j^{\text{eq}}, \quad (80a)$$

$$\sum_j \mathbf{c}_j \mathbf{c}_j f_j^{\text{eq}} = c_s^2 \rho \mathbf{I} + \rho \mathbf{u} \mathbf{u}, \quad \sum_j \mathbf{c}_j \mathbf{c}_j \mathbf{c}_j f_j^{\text{eq}} = c_s^2 \rho \Delta \cdot \mathbf{u}, \quad (80b)$$

$$\sum_j G_j = 0, \quad \sum_j \mathbf{c}_j G_j = \mathbf{M}_{1G} = 0, \quad \sum_j \mathbf{c}_j \mathbf{c}_j G_j = \mathbf{M}_{2G}, \quad (80c)$$

$$\sum_j F_j = \bar{S}, \quad \sum_j \mathbf{c}_j F_j = \mathbf{M}_{1F} = \bar{\mathbf{F}}, \quad \sum_j \mathbf{c}_j \mathbf{c}_j F_j = \mathbf{M}_{2F} = 0, \quad (80d)$$

$$\sum_j \mathbf{e}_j \Lambda_{jk} = s_0 \mathbf{e}_k, \quad \sum_j \mathbf{c}_j \Lambda_{jk} = \mathbf{S}_{10} \mathbf{e}_k + \mathbf{S}_1 \mathbf{c}_k, \quad (80e)$$

$$\begin{aligned} \sum_j \mathbf{c}_j \mathbf{c}_j \Lambda_{jk} &= \sum_j \left[\mathbf{c}_j \mathbf{c}_j - \frac{1}{d} (\mathbf{c}_j \cdot \mathbf{c}_j) \mathbf{I} \right] \Lambda_{jk} + \sum_j \frac{1}{d} [(\mathbf{c}_j \cdot \mathbf{c}_j) \mathbf{I}] \Lambda_{jk} \\ &= \mathbf{S}_{20} \mathbf{e}_k + \mathbf{S}_{21} \mathbf{c}_k + S_{2s} \left[\mathbf{c}_k \mathbf{c}_k - \frac{1}{d} (\mathbf{c}_k \cdot \mathbf{c}_k) \mathbf{I} \right] + \frac{S_{2b}}{d} (\mathbf{c}_k \cdot \mathbf{c}_k) \mathbf{I}, \end{aligned} \quad (80f)$$

where $\mathbf{M}_{2G}$ is a second-order tensor to be determined below, $\Delta_{\alpha\beta\gamma\theta} = \delta_{\alpha\beta}\delta_{\gamma\theta} + \delta_{\alpha\gamma}\delta_{\beta\theta} + \delta_{\alpha\theta}\delta_{\beta\gamma}$, $\mathbf{S}_1$ is an invertible $d \times d$ relaxation matrix, $\mathbf{S}_{20}$ and $\mathbf{S}_{21}$ are two $d^2 \times 1$ and $d^2 \times d$ matrices, and S_{2s} and S_{2b} are the relaxation parameters corresponding to the second-order moments.

1. Derivation of NSEs through the DTE method

Due to the similarity between the CE analysis and some other methods, here we only use the DTE method, i.e.,

Eqs. (24a) and (27), to recover NSEs. With the help of Eqs. (29) and (30), we can rewrite Eqs. (24a) and (27) as

$$D_j f_j^{\text{eq}} = \frac{1}{\Delta t} g_j + G_j + F_j + O(\Delta t), \quad (81a)$$

$$\begin{aligned} D_j f_j^{\text{eq}} - D_j \bar{\Lambda}_{jk} g_k &= \frac{1}{\Delta t} g_j + G_j + F_j + \frac{\Delta t}{2} [(\gamma - 1) \mathbf{c}_j \\ &\quad \cdot \nabla F_j - D_j G_j] + O(\Delta t^2). \end{aligned} \quad (81b)$$

On the other hand, from Eqs. (29) and (80) we have

$$\sum_j \mathbf{e}_j g_j = -s_0 \sum_k \mathbf{e}_k f_k^{\text{ne}} = 0, \quad \sum_j \mathbf{c}_j g_j = -\mathbf{S}_1 \sum_k \mathbf{c}_k f_k^{\text{ne}} = 0, \quad (82a)$$

$$\sum_j \mathbf{e}_j \bar{\Lambda}_{jk} = (1/s_0 - 1/2) \mathbf{e}_k, \quad \sum_j \mathbf{c}_j \bar{\Lambda}_{jk} = -\frac{1}{s_0} \mathbf{S}_1^{-1} \mathbf{S}_{10} \mathbf{e}_k + (\mathbf{S}_1^{-1} - \mathbf{I}/2) \mathbf{c}_k, \quad (82b)$$

$$\begin{aligned} \sum_j \mathbf{c}_j \mathbf{c}_j \bar{\Lambda}_{jk} &= \sum_j \left[\mathbf{c}_j \mathbf{c}_j - \frac{1}{d} (\mathbf{c}_j \cdot \mathbf{c}_j) \mathbf{I} \right] \bar{\Lambda}_{jk} + \sum_j \frac{1}{d} [(\mathbf{c}_j \cdot \mathbf{c}_j) \mathbf{I}] \bar{\Lambda}_{jk} \\ &= -\frac{1}{s_0} \left(\frac{1}{S_{2s}} + \frac{1}{S_{2b}} \right) (\mathbf{S}_{20} - \mathbf{S}_{21} \mathbf{S}_1^{-1} \mathbf{S}_{10}) \mathbf{e}_k - \left(\frac{1}{S_{2s}} + \frac{1}{S_{2b}} \right) \mathbf{S}_{21} \mathbf{S}_1^{-1} \mathbf{c}_k \\ &\quad + \left(\frac{1}{S_{2s}} - \frac{1}{2} \right) \left[\mathbf{c}_k \mathbf{c}_k - \frac{1}{d} (\mathbf{c}_k \cdot \mathbf{c}_k) \mathbf{I} \right] + \left(\frac{1}{S_{2b}} - \frac{1}{2} \right) \frac{1}{d} (\mathbf{c}_k \cdot \mathbf{c}_k) \mathbf{I} \\ &= -\frac{1}{s_0} \left(\frac{1}{S_{2s}} + \frac{1}{S_{2b}} \right) (\mathbf{S}_{20} - \mathbf{S}_{21} \mathbf{S}_1^{-1} \mathbf{S}_{10}) \mathbf{e}_k - \left(\frac{1}{S_{2s}} + \frac{1}{S_{2b}} \right) \mathbf{S}_{21} \mathbf{S}_1^{-1} \mathbf{c}_k \\ &\quad + \left(\frac{1}{S_{2s}} - \frac{1}{2} \right) \mathbf{c}_k \mathbf{c}_k + \left(\frac{1}{S_{2b}} - \frac{1}{2} \right) \frac{1}{d} (\mathbf{c}_k \cdot \mathbf{c}_k) \mathbf{I}, \end{aligned} \quad (82c)$$

and

$$\sum_j \mathbf{c}_{j\alpha} D_j \bar{\mathbf{A}}_{jk} g_k = \partial_t \sum_j \mathbf{c}_{j\alpha} \bar{\mathbf{A}}_{jk} g_k + \nabla_\beta \cdot \sum_j \mathbf{c}_{j\alpha} \mathbf{c}_{j\beta} \bar{\mathbf{A}}_{jk} g_k + \left(\frac{1}{S_{2b}} - \frac{1}{S_{2s}} \right) \frac{\delta_{\alpha\beta}}{d} \mathbf{c}_{k\gamma} \mathbf{c}_{k\gamma} \Big] g_k. \quad (83)$$

$$= \partial_t (\mathbf{S}_1^{-1} - \mathbf{I}/2) \sum_k \mathbf{c}_{k\alpha} g_k$$

$$+ \nabla_\beta \cdot \sum_j \mathbf{c}_{j\alpha} \mathbf{c}_{j\beta} \bar{\mathbf{A}}_{jk} g_k$$

$$= \nabla_\beta \cdot \sum_j \mathbf{c}_{j\alpha} \mathbf{c}_{j\beta} \bar{\mathbf{A}}_{jk} g_k$$

$$= \nabla_\beta \cdot \sum_k \left[\left(\frac{1}{S_{2s}} - \frac{1}{2} \right) \mathbf{c}_{k\alpha} \mathbf{c}_{k\beta} \right]$$

Summing Eq. (81b) and adopting Eqs. (80) and (82), we can obtain

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = \frac{\Delta t}{2} (\gamma - 1) \nabla \cdot \bar{\mathbf{F}} + \bar{S} + O(\Delta t^2). \quad (84)$$

If we take $\gamma = 1$, then the continuity equation (78a) is recovered correctly to the order of $O(\Delta t^2)$.

Multiplying $\mathbf{c}_j$ on both sides of Eq. (81), and through a summation over j , we have

$$\partial_t (\rho \mathbf{u}) + \nabla \cdot (c_s^2 \rho \mathbf{I} + \rho \mathbf{u} \mathbf{u}) = \bar{\mathbf{F}} + O(\Delta t), \quad (85a)$$

$$\begin{aligned} \partial_t (\rho \mathbf{u}_\alpha) + \nabla_\beta \cdot (c_s^2 \rho \delta_{\alpha\beta} + \rho \mathbf{u}_\alpha \mathbf{u}_\beta) - \nabla_\beta \cdot \left[\left(\frac{1}{S_{2s}} - \frac{1}{2} \right) \mathbf{c}_{k\alpha} \mathbf{c}_{k\beta} + \left(\frac{1}{S_{2b}} - \frac{1}{S_{2s}} \right) \frac{\delta_{\alpha\beta}}{d} \mathbf{c}_{k\gamma} \mathbf{c}_{k\gamma} \right] g_k \\ = \bar{\mathbf{F}}_\alpha - \frac{\Delta t}{2} \nabla_\beta \cdot \mathbf{M}_{2G,\alpha\beta} + O(\Delta t^2), \end{aligned} \quad (85b)$$

where Eqs. (80), (82), and (83) have been used.

Additionally, from Eqs. (80) and (81) we get

$$\begin{aligned} \sum_k \mathbf{c}_{k\alpha} \mathbf{c}_{k\beta} g_k &= \Delta t \sum_k \mathbf{c}_{k\alpha} \mathbf{c}_{k\beta} (D_k f_k^{\text{eq}} - G_k - F_k) + O(\Delta t^2) \\ &= \Delta t \left(\partial_t \sum_k \mathbf{c}_{k\alpha} \mathbf{c}_{k\beta} f_k^{\text{eq}} + \nabla_\gamma \cdot \sum_k \mathbf{c}_{k\alpha} \mathbf{c}_{k\beta} \mathbf{c}_{k\gamma} f_k^{\text{eq}} - \mathbf{M}_{2G,\alpha\beta} \right) + O(\Delta t^2) \\ &= \Delta t [\partial_t (c_s^2 \rho \delta_{\alpha\beta} + \rho \mathbf{u}_\alpha \mathbf{u}_\beta) + \nabla_\gamma \cdot (c_s^2 \rho \Delta_{\alpha\beta\gamma\theta} \cdot \mathbf{u}_\theta) - \mathbf{M}_{2G,\alpha\beta}] + O(\Delta t^2) \\ &= \Delta t [\partial_t (c_s^2 \rho \delta_{\alpha\beta} + \rho \mathbf{u}_\alpha \mathbf{u}_\beta) + \nabla_\gamma \cdot c_s^2 \rho (\mathbf{u}_\alpha \delta_{\beta\gamma} + \mathbf{u}_\beta \delta_{\alpha\gamma} + \mathbf{u}_\gamma \delta_{\alpha\beta}) - \mathbf{M}_{2G,\alpha\beta}] + O(\Delta t^2). \end{aligned} \quad (86)$$

Based on the following equations:

$$\partial_t (\rho \mathbf{u} \mathbf{u}) = \mathbf{u} \bar{\mathbf{F}} + \bar{\mathbf{F}} \mathbf{u} - c_s^2 [\mathbf{u} \nabla \rho + (\mathbf{u} \nabla \rho)^T] - \nabla \cdot (\rho \mathbf{u} \mathbf{u} \mathbf{u}) - \mathbf{u} \mathbf{u} \bar{S} + O(\Delta t), \quad (87a)$$

$$\nabla (c_s^2 \rho \mathbf{u}) + [\nabla (c_s^2 \rho \mathbf{u})]^T = c_s^2 \rho [\nabla \mathbf{u} + (\nabla \mathbf{u})^T] + c_s^2 [\mathbf{u} \nabla \rho + (\mathbf{u} \nabla \rho)^T], \quad (87b)$$

we can rewrite Eq. (86) as

$$\sum_k \mathbf{c}_{k\alpha} \mathbf{c}_{k\beta} g_k = \Delta t [\rho c_s^2 (\partial_\alpha \mathbf{u}_\beta + \partial_\beta \mathbf{u}_\alpha) + \mathbf{u}_\alpha \bar{\mathbf{F}}_\beta + \bar{\mathbf{F}}_\alpha \mathbf{u}_\beta + (c_s^2 \delta_{\alpha\beta} - \mathbf{u}_\alpha \mathbf{u}_\beta) \bar{S} - \mathbf{M}_{2G,\alpha\beta}] + O(\Delta t^2 + \Delta t \text{Ma}^3), \quad (88a)$$

$$\sum_k \mathbf{c}_{k\gamma} \mathbf{c}_{k\gamma} g_k \delta_{\alpha\beta} = \Delta t [2\rho c_s^2 \partial_\gamma \mathbf{u}_\gamma + 2\mathbf{u}_\gamma \bar{\mathbf{F}}_\gamma + (dc_s^2 - \mathbf{u}_\gamma \mathbf{u}_\gamma) \bar{S} - \mathbf{M}_{2G,\gamma\gamma}] \delta_{\alpha\beta} + O(\Delta t^2 + \Delta t \text{Ma}^3), \quad (88b)$$

where Ma is the Mach number, and the relation $\nabla_\gamma \cdot (\rho \mathbf{u}_\alpha \mathbf{u}_\beta \mathbf{u}_\gamma) = O(\text{Ma}^3)$ is used to derive above Eq. (88). Substituting Eq. (88) into Eq. (85b) and using Eq. (84), we can obtain

$$\begin{aligned} \partial_t (\rho \mathbf{u}_\alpha) + \nabla_\beta \cdot (c_s^2 \rho \delta_{\alpha\beta} + \rho \mathbf{u}_\alpha \mathbf{u}_\beta) &= \nabla_\beta \cdot \left[\Delta t \left(\frac{1}{S_{2s}} - \frac{1}{2} \right) c_s^2 \rho \left(\nabla_\alpha \mathbf{u}_\beta + \nabla_\beta \mathbf{u}_\alpha - \frac{2}{d} \nabla_\gamma \mathbf{u}_\gamma \delta_{\alpha\beta} \right) \right. \\ &\quad \left. + \frac{2}{d} \Delta t \left(\frac{1}{S_{2b}} - \frac{1}{2} \right) c_s^2 \rho \nabla_\gamma \mathbf{u}_\gamma \delta_{\alpha\beta} \right] + \bar{\mathbf{F}}_\alpha + \Delta t \nabla_\beta \cdot (RH_2)_{\alpha\beta} + O(\Delta t^2 + \Delta t \text{Ma}^3), \end{aligned} \quad (89)$$

where

$$\begin{aligned}
 (RH_2)_{\alpha\beta} &= \left(\frac{1}{S_{2s}} - \frac{1}{2}\right) \left[\bar{\mathbf{F}}_\alpha \mathbf{u}_\beta + \mathbf{u}_\alpha \bar{\mathbf{F}}_\beta - \frac{2}{d} \mathbf{u}_\gamma \bar{\mathbf{F}}_\gamma \delta_{\alpha\beta} - \left(\mathbf{u}_\alpha \mathbf{u}_\beta - \frac{1}{d} \mathbf{u}_\gamma^2 \delta_{\alpha\beta} \right) \bar{\mathbf{S}} - \mathbf{M}_{2G,\alpha\beta} + \frac{1}{d} \mathbf{M}_{2G,\gamma\gamma} \delta_{\alpha\beta} \right] \\
 &\quad + \left(\frac{1}{S_{2b}} - \frac{1}{2}\right) \left[\frac{2}{d} \mathbf{u}_\gamma \bar{\mathbf{F}}_\gamma \delta_{\alpha\beta} + \left(c_s^2 - \frac{1}{d} \mathbf{u}_\gamma^2 \right) \delta_{\alpha\beta} \bar{\mathbf{S}} - \frac{1}{d} \mathbf{M}_{2G,\gamma\gamma} \delta_{\alpha\beta} \right] - \frac{1}{2} \mathbf{M}_{2G,\alpha\beta} \\
 &= \left(\frac{1}{S_{2s}} - \frac{1}{2}\right) \left[\bar{\mathbf{F}}_\alpha \mathbf{u}_\beta + \mathbf{u}_\alpha \bar{\mathbf{F}}_\beta - (\mathbf{u}_\alpha \mathbf{u}_\beta - c_s^2 \delta_{\alpha\beta}) \bar{\mathbf{S}} - \mathbf{M}_{2G,\alpha\beta} \right] \\
 &\quad + \left(\frac{1}{S_{2b}} - \frac{1}{S_{2s}}\right) \left[\frac{2}{d} \mathbf{u}_\gamma \bar{\mathbf{F}}_\gamma + \left(c_s^2 - \frac{1}{d} \mathbf{u}_\gamma^2 \right) \bar{\mathbf{S}} - \frac{1}{d} \mathbf{M}_{2G,\gamma\gamma} \right] \delta_{\alpha\beta} - \frac{1}{2} \mathbf{M}_{2G,\alpha\beta}.
 \end{aligned} \tag{90}$$

To obtain the correct NSEs, $(RH_2)_{\alpha\beta} = 0$, which gives rise to following equations:

$$\mathbf{M}_{2G,\gamma\gamma} = \left(1 - \frac{S_{2b}}{2}\right) [2\mathbf{u}_\gamma \bar{\mathbf{F}}_\gamma + (dc_s^2 - \mathbf{u}_\gamma \mathbf{u}_\gamma) \bar{\mathbf{S}}], \tag{91a}$$

$$\begin{aligned}
 \mathbf{M}_{2G,\alpha\beta} &= \left(1 - \frac{S_{2s}}{2}\right) \left[\bar{\mathbf{F}}_\alpha \mathbf{u}_\beta + \bar{\mathbf{F}}_\beta \mathbf{u}_\alpha + (c_s^2 \delta_{\alpha\beta} - \mathbf{u}_\alpha \mathbf{u}_\beta) \bar{\mathbf{S}} - \frac{2}{d} \mathbf{u}_\gamma \bar{\mathbf{F}}_\gamma \delta_{\alpha\beta} - \left(c_s^2 - \frac{\mathbf{u}_\gamma \mathbf{u}_\gamma}{d} \right) \bar{\mathbf{S}} \delta_{\alpha\beta} \right] \\
 &\quad + \left(1 - \frac{S_{2b}}{2}\right) \left[\frac{2}{d} \mathbf{u}_\gamma \bar{\mathbf{F}}_\gamma \delta_{\alpha\beta} + \left(c_s^2 - \frac{\mathbf{u}_\gamma \mathbf{u}_\gamma}{d} \right) \bar{\mathbf{S}} \delta_{\alpha\beta} \right].
 \end{aligned} \tag{91b}$$

Introduce the matrix $\mathbf{W} = \bar{\mathbf{F}}\mathbf{u} + \mathbf{u}\bar{\mathbf{F}} + (c_s^2 \mathbf{I} - \mathbf{u}\mathbf{u})\bar{\mathbf{S}}$, and then Eq. (91b) can be written as

$$\mathbf{M}_{2G} = \left(1 - \frac{S_{2s}}{2}\right) \left[\mathbf{W} - \frac{1}{d} \text{tr}(\mathbf{W}) \mathbf{I} \right] + \frac{1}{d} \left(1 - \frac{S_{2b}}{2}\right) \text{tr}(\mathbf{W}) \mathbf{I}, \tag{92}$$

where $\text{tr}(\mathbf{W})$ is the trace of the matrix $\mathbf{W}$. We note that when $\bar{\mathbf{S}} = 0$ and $S_{2s} = S_{2b}$, Eq. (92) would be the same as the second-order moment of the forcing term distribution function in the previous work [81,82].

Based on $\mathbf{M}_{2G}$ defined by Eq. (92), we can write Eq. (89) in the following form:

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot (c_s^2 \rho \mathbf{I} + \rho \mathbf{u}\mathbf{u}) = \nabla \cdot \mu \left[\nabla \mathbf{u} + (\nabla \mathbf{u})^T - \frac{2}{d} (\nabla \cdot \mathbf{u}) \mathbf{I} \right] + \nabla(\mu_b \nabla \cdot \mathbf{u}) + \bar{\mathbf{F}} + O(\Delta t^2 + \Delta t \text{Ma}^3). \tag{93}$$

When $p = \rho c_s^2$ and the truncation error $O(\Delta t^2 + \Delta t \text{Ma}^3)$ is neglected, Eq. (93) would reduce to the momentum equation (78b) with the following dynamic and bulk viscosities:

$$\mu = \left(\frac{1}{S_{2s}} - \frac{1}{2}\right) \rho c_s^2 \Delta t, \quad \mu_b = \frac{2}{d} \left(\frac{1}{S_{2b}} - \frac{1}{2}\right) \rho c_s^2 \Delta t. \tag{94}$$

It should be noted that although the present MRT-LB model can recover the compressible NSEs [Eq. (78)] correctly, it is only suitable for the incompressible or weakly compressible fluid flows. The reason is twofold. The first is that in the present LB model, the discrete equilibrium function f_i^{eq} [see Eq. (98a)] is only a second-order approximation to the continuous Maxwell-Boltzmann distribution where the assumption of low velocity or small Mach number (the small Mach number is equivalent to incompressible limit [83]) is adopted, and the high-order terms in velocity or Mach number have been neglected. The other is that in the derivation of the NSEs, the third-order term in Mach number has also been omitted, as discussed above. Finally, we also present a local scheme to calculate shear stress or strain rate tensor in the framework of LB method [66,84,85]. For incompressible fluid flows, the term $O(\Delta t \text{Ma}^3)$ in Eq. (88a) can be neglected, and then we can obtain the following expression of strain rate tensor with a second-order accuracy in time:

$$\begin{aligned}
 \frac{\partial_\alpha \mathbf{u}_\beta + \partial_\beta \mathbf{u}_\alpha}{2} &= \frac{1}{2\rho c_s^2 \Delta t} \sum_k \mathbf{c}_{k\alpha} \mathbf{c}_{k\beta} g_k - \frac{1}{2\rho c_s^2} [\mathbf{u}_\alpha \bar{\mathbf{F}}_\beta + \bar{\mathbf{F}}_\alpha \mathbf{u}_\beta + (c_s^2 \delta_{\alpha\beta} - \mathbf{u}_\alpha \mathbf{u}_\beta) \bar{\mathbf{S}} - \mathbf{M}_{2G,\alpha\beta}] \\
 &= \frac{2 - S_{2s}}{4S_{2s}\mu} \sum_k \left[-S_{2s} \mathbf{c}_{k\alpha} \mathbf{c}_{k\beta} + (S_{2b} - S_{2s}) \frac{\delta_{\alpha\beta}}{d} \mathbf{c}_{k\gamma} \mathbf{c}_{k\gamma} \right] f_k^{\text{ne}} \\
 &\quad - \frac{1}{2\rho c_s^2} [\mathbf{u}_\alpha \bar{\mathbf{F}}_\beta + \bar{\mathbf{F}}_\alpha \mathbf{u}_\beta + (c_s^2 \delta_{\alpha\beta} - \mathbf{u}_\alpha \mathbf{u}_\beta) \bar{\mathbf{S}} - \mathbf{M}_{2G,\alpha\beta}],
 \end{aligned} \tag{95}$$

where Eq. (87a) has been used. Based on Eq. (95), one can also give a local scheme for shear stress with a second-order accuracy in time,

$$\begin{aligned} \mu(\partial_\alpha \mathbf{u}_\beta + \partial_\beta \mathbf{u}_\alpha) = & \left(\frac{1}{S_{2s}} - \frac{1}{2} \right) \sum_k \left[-S_{2s} \mathbf{c}_{k\alpha} \mathbf{c}_{k\beta} + (S_{2b} - S_{2s}) \frac{\delta_{\alpha\beta}}{d} \mathbf{c}_{k\gamma} \mathbf{c}_{k\gamma} \right] f_k^{\text{ne}} \\ & - \left(\frac{1}{S_{2s}} - \frac{1}{2} \right) \Delta t \left[\mathbf{u}_\alpha \bar{\mathbf{F}}_\beta + \bar{\mathbf{F}}_\alpha \mathbf{u}_\beta + (c_s^2 \delta_{\alpha\beta} - \mathbf{u}_\alpha \mathbf{u}_\beta) \bar{S} - \mathbf{M}_{2G, \alpha\beta} \right]. \end{aligned} \quad (96)$$

We note that Eqs. (95) and (96) are the same as the results in Ref. [66] once $\bar{S} = 0$.

Remark: For the steady-state problems govern by the NSEs (78), one can derive the relation $\bar{S} = \nabla \cdot (\rho \mathbf{u}) \sim O(\text{Ma})$ from Eq. (78a). Based on this relation, we can rewrite Eq. (87a) as

$$\begin{aligned} \mathbf{u} \bar{\mathbf{F}} + \bar{\mathbf{F}} \mathbf{u} = & c_s^2 [\mathbf{u} \nabla \rho + (\mathbf{u} \nabla \rho)^T] + \nabla \cdot (\rho \mathbf{u} \mathbf{u} \mathbf{u}) + \mathbf{u} \mathbf{u} \bar{S} \\ & + O(\Delta t) \sim O(\Delta t + \text{Ma}^3), \end{aligned} \quad (97)$$

where the term related to time derivative has been neglected. From above equation, we can find that the term $\mathbf{u} \bar{\mathbf{F}} + \bar{\mathbf{F}} \mathbf{u}$ can be omitted in the derivation of the steady-state NSEs, and in this case, the present results are also consistent with those in the previous work [75].

2. The equilibrium, auxiliary, and source distribution functions of the MRT-LB method for NSEs

From the above discussion, one can clearly observe that to recover the macroscopic NSEs (78) from the MRT-LB method (1), the equilibrium, auxiliary, and source distribution functions should satisfy some necessary requirements, as depicted by Eq. (80). For a general DdQq lattice model, the explicit expressions of f_j^{eq} , G_j , and F_j can be given by

$$f_j^{\text{eq}} = \omega_j \rho \left[1 + \frac{\mathbf{c}_j \cdot \mathbf{u}}{c_s^2} + \frac{\mathbf{u} \mathbf{u} : (\mathbf{c}_j \mathbf{c}_j - c_s^2 \mathbf{I})}{2c_s^4} \right], \quad (98a)$$

$$G_j = \omega_j \frac{\mathbf{M}_{2G} : (\mathbf{c}_j \mathbf{c}_j - c_s^2 \mathbf{I})}{2c_s^4}, \quad (98b)$$

$$F_j = \omega_j \left(\bar{S} + \frac{\mathbf{c}_j \cdot \bar{\mathbf{F}}}{c_s^2} \right) - \frac{\bar{S} c_s^2 \mathbf{I} : (\mathbf{c}_j \mathbf{c}_j - c_s^2 \mathbf{I})}{2c_s^4}, \quad (98c)$$

where $\mathbf{M}_{2G}$ is determined by Eq. (92).

From the discussion in this section, one can find that the NSEs and NCDEs can be recovered correctly from the general MRT-LB method, and in other words, the present MRT-LB method not only can be used to study fluid flow problems governed by the NSEs but also can be adopted to investigate the heat and mass transport problems depicted by the coupled NCDEs and NSEs.

V. COLLISION MATRIX AND SOME SPECIAL CASES OF THE MRT-LB METHOD

When the discrete velocity set $V_q = \{\mathbf{c}_j, 0 \leq j \leq q-1\}$ is given, we can construct different forms of collision matrix Λ in MRT-LB method.

In general, the collision matrix Λ has the following form:

$$\Lambda = \mathbf{M}^{-1} \mathbf{S} \mathbf{M}, \quad (99a)$$

$$\mathbf{S} = (\mathbf{S}_{kj}), \quad \mathbf{S}_{kj} = 0 \quad (k < j), \quad \mathbf{S}_{kk} = \mathbf{S}_k, \quad (99b)$$

where the rows of transformation matrix $\mathbf{M}$ are made up of discrete velocities in V_q , $\mathbf{S}$ is a block-lower-triangle matrix, and $\mathbf{S}_k \in R^{n_k \times n_k}$ is a relaxation matrix corresponding to the k th ($0 \leq k \leq m$) order moment of discrete velocity. Denote $\mathbf{M}$ with the following form:

$$\mathbf{M} = (\mathbf{M}_0^T, \mathbf{M}_1^T, \mathbf{M}_2^T, \dots, \mathbf{M}_m^T)^T, \quad \mathbf{M}_k \in R^{n_k \times q}, \quad 0 \leq k \leq m, \quad (100)$$

then we have

$$\mathbf{M}_k \Lambda = \sum_{j=0}^k \mathbf{S}_{kj} \mathbf{M}_j, \quad 0 \leq k \leq m, \quad (101)$$

with

$$\mathbf{M}_0 = \mathbf{e} = (1, 1, \dots, 1), \quad \mathbf{M}_1 = \mathbf{E} = (\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_{q-1}),$$

$$\mathbf{M}_2 = \begin{pmatrix} \mathbf{M}_2^{(1)} \\ \mathbf{M}_2^{(2)} \end{pmatrix}. \quad (102a)$$

$$S_0 \in R, \quad S_1 \in R^{d \times d}, \quad S_2 \in R^{\bar{d} \times \bar{d}}, \quad \bar{d} = d(d+1)/2, \quad (102b)$$

where $\mathbf{M}_2^{(1)} = (\mathbf{c}_{0\alpha} \mathbf{c}_{0\alpha}, \mathbf{c}_{1\alpha} \mathbf{c}_{1\alpha}, \dots, \mathbf{c}_{q-1\alpha} \mathbf{c}_{q-1\alpha})$ and $\mathbf{M}_2^{(2)} = (\mathbf{c}_{0\alpha} \mathbf{c}_{0\beta}, \mathbf{c}_{1\alpha} \mathbf{c}_{1\beta}, \dots, \mathbf{c}_{q-1\alpha} \mathbf{c}_{q-1\beta})_{\alpha < \beta}$.

For a particular case $\mathbf{S}_{kj} \quad (k > j) = 0$, one can obtain

$$\mathbf{S} = \text{diag}(\mathbf{S}_0, \mathbf{S}_1, \dots, \mathbf{S}_m), \quad (103)$$

and Eq. (101) would reduce to

$$\mathbf{M}_k \Lambda = \mathbf{S}_k \mathbf{M}_k, \quad 0 \leq k \leq m, \quad (104)$$

which can be used to derive the commonly used MRT-LB model [4,5].

Based on Eq. (80f), we can determine the relaxation matrix $\mathbf{S}_2$ as

$$\mathbf{S}_2 = S_{2s} \mathbf{I} + \frac{(S_{2b} - S_{2s})}{d} \begin{pmatrix} \mathbb{E}_d & 0 \\ 0 & 0 \end{pmatrix}, \quad (105)$$

where $\mathbb{E}_d$ is a $d \times d$ matrix with all elements $(\mathbb{E}_d)_{ij} = 1$.

We now point out that some commonly used LB models in the literature can be obtained from our MRT-LB method when the matrices Λ , $\mathbf{M}$, and $\mathbf{S}$ are specified with some special forms.

(1) Lattice BGK or SRT-LB model [6,7]:

$$\Lambda = \mathbf{S} = \frac{1}{\tau} \mathbf{I}, \quad (106)$$

where the parameter τ is the relaxation time in the lattice BGK model [2,7].

(2) **TRT-LB model** [19–21]:

$$\mathbf{S}_{2k} = s^+ \mathbf{I}_{2k}, \mathbf{S}_{2k+1} = s^- \mathbf{I}_{2k+1}, 0 \leq 2k, 2k+1 \leq m, \quad (107)$$

where $\mathbf{S}_{2k}$ and $\mathbf{S}_{2k+1}$ are the relaxation matrices corresponding to the $2k$ and $(2k+1)$ th order moments of discrete velocity.

Let

$$q-1 = 2\bar{m}, \quad (108a)$$

$$\begin{aligned} \mathbf{V}_q &= \{\mathbf{c}_j, j = 0, 1, \dots, 2\bar{m}\} \\ &= \{\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_{\bar{m}}, -\mathbf{c}_1, \dots, -\mathbf{c}_{\bar{m}}\}, \end{aligned} \quad (108b)$$

the collision matrix $\mathbf{\Lambda}$ can be written as

$$\mathbf{\Lambda} = \begin{bmatrix} \omega + \bar{\omega} & 0 & 0 \\ 0 & \omega \mathbf{I}_{\bar{m}} & \bar{\omega} \mathbf{I}_{\bar{m}} \\ 0 & \bar{\omega} \mathbf{I}_{\bar{m}} & \omega \mathbf{I}_{\bar{m}} \end{bmatrix}, \quad (109)$$

where $\omega = (s^+ + s^-)/2$ and $\bar{\omega} = (s^+ - s^-)/2$ with s^+ and s^- being the relaxation parameters corresponding to the symmetric and antisymmetric parts of distribution functions [19] or, equivalently, $s^+ = \omega + \bar{\omega}$, $s^- = \omega - \bar{\omega}$. Actually, if we substitute the present collision matrix (109) into Eq. (1), then one can derive the following TRT-LB model:

$$\begin{aligned} f_j(\mathbf{x} + \mathbf{c}_j \Delta t, t + \Delta t) \\ = f_j(\mathbf{x}, t) - \omega f_j^{\text{ne}}(\mathbf{x}, t) - \bar{\omega} f_j^{\text{eq}}(\mathbf{x}, t) \\ + \Delta t \left[G_j(\mathbf{x}, t) + F_j(\mathbf{x}, t) + \frac{\Delta t}{2} \bar{D}_j F_j(\mathbf{x}, t) \right], \end{aligned} \quad (110)$$

where $\bar{j}$ is the opposite direction of j , $\mathbf{c}_{\bar{j}} = -\mathbf{c}_j$. After some algebraic manipulations, the above equation can be equivalently

written as [19]

$$\begin{aligned} f_j(\mathbf{x} + \mathbf{c}_j \Delta t, t + \Delta t) \\ = f_j(\mathbf{x}, t) - s^+ f_j^{\text{ne},+}(\mathbf{x}, t) - s^- f_j^{\text{ne},-}(\mathbf{x}, t) \\ + \Delta t \left[G_j(\mathbf{x}, t) + F_j(\mathbf{x}, t) + \frac{\Delta t}{2} \bar{D}_j F_j(\mathbf{x}, t) \right], \end{aligned} \quad (111)$$

where $f_j^{\text{ne},+} = f_j^+ - f_j^{\text{eq},+}$ and $f_j^{\text{ne},-} = f_j^- - f_j^{\text{eq},-}$ with $f_j^\pm = (f_j \pm f_{\bar{j}})/2$ and $f_j^{\text{eq},\pm} = (f_j^{\text{eq}} \pm f_{\bar{j}}^{\text{eq}})/2$. From Eqs. (110) and (111), one can also find that compared to the lattice BGK model, there is no apparent computational cost increased in the TRT-LB model.

In addition, we also note that if $\mathbf{c}_0 = \mathbf{0}$ is not contained in the discrete velocity set $\mathbf{V}_q$, then the matrix $\mathbf{\Lambda}$ in Eq. (109) would be given by

$$\mathbf{\Lambda} = \begin{bmatrix} \omega \mathbf{I}_{\bar{m}} & \bar{\omega} \mathbf{I}_{\bar{m}} \\ \bar{\omega} \mathbf{I}_{\bar{m}} & \omega \mathbf{I}_{\bar{m}} \end{bmatrix}, \quad (112)$$

Now let us focus on how to derive the REs [Eqs. (35) and (36)] of the TRT model. Based on the collision matrix of the TRT model [see Eq. (109)], we can write $\bar{\mathbf{\Lambda}}$ as

$$\bar{\mathbf{\Lambda}} = \mathbf{\Lambda}^{-1} - \mathbf{I}/2 = \begin{bmatrix} \lambda^+ + \lambda^- & 0 & 0 \\ 0 & \lambda^+ \mathbf{I}_{\bar{m}} & \lambda^- \mathbf{I}_{\bar{m}} \\ 0 & \lambda^- \mathbf{I}_{\bar{m}} & \lambda^+ \mathbf{I}_{\bar{m}} \end{bmatrix}, \quad (113)$$

where $\lambda^+ = (\lambda_e + \lambda_o)/2$, $\lambda^- = (\lambda_e - \lambda_o)/2$ with $\lambda_e = 1/s^+ - 1/2$ and $\lambda_o = 1/s^- - 1/2$. If we substitute Eq. (113) into Eqs. (35) and (36), then one can obtain

$$[(\Delta_j^2 - \Delta_t^2)(\bar{f}_j^{\text{eq}} - \lambda^+ \bar{g}_j - \lambda^- \bar{g}_{\bar{j}}) - (\bar{\Delta}_j - \bar{\Delta}_t) \bar{g}_{\bar{j}}](\mathbf{x}, t) = 0, \quad (114a)$$

$$\left[(\bar{\Delta}_j + \bar{\Delta}_t)(\bar{f}_j^{\text{eq}} - \lambda^+ \bar{g}_j - \lambda^- \bar{g}_{\bar{j}}) - \frac{1}{4}(\Delta_j^2 + \Delta_t^2) \bar{g}_{\bar{j}} \right](\mathbf{x}, t) = \bar{g}_{\bar{j}}(\mathbf{x}, t), \quad (114b)$$

$$[(\Delta_j^2 - \Delta_t^2)(\bar{f}_j^{\text{eq}} - \lambda^+ \bar{g}_j - \lambda^- \bar{g}_{\bar{j}}) + (\bar{\Delta}_j + \bar{\Delta}_t) \bar{g}_{\bar{j}}](\mathbf{x}, t) = 0, \quad (115a)$$

$$\left[(-\bar{\Delta}_j + \bar{\Delta}_t)(\bar{f}_j^{\text{eq}} - \lambda^+ \bar{g}_j - \lambda^- \bar{g}_{\bar{j}}) - \frac{1}{4}(\Delta_j^2 + \Delta_t^2) \bar{g}_{\bar{j}} \right](\mathbf{x}, t) = \bar{g}_{\bar{j}}(\mathbf{x}, t). \quad (115b)$$

Substituting Eqs. (114a) and (115a) into Eqs. (114b) and (115b) yields

$$\bar{g}_j = (\bar{\Delta}_j + \bar{\Delta}_t) \bar{f}_j^{\text{eq}} - (\Delta_j^2 - \Delta_t^2)(\lambda^+ \bar{f}_j^{\text{eq}} - \lambda^- \bar{f}_{\bar{j}}^{\text{eq}}) + (\Delta_j^2 - \Delta_t^2) \left(\lambda_e \lambda_o - \frac{1}{4} \right) \bar{g}_j - 2\bar{\Delta}_t \lambda^+ \bar{g}_j - \frac{1}{2} \Delta_t^2 \bar{g}_j, \quad (116a)$$

$$\bar{g}_{\bar{j}} = (-\bar{\Delta}_{\bar{j}} + \bar{\Delta}_t) \bar{f}_{\bar{j}}^{\text{eq}} - (\Delta_j^2 - \Delta_t^2)(\lambda^+ \bar{f}_j^{\text{eq}} - \lambda^- \bar{f}_{\bar{j}}^{\text{eq}}) + (\Delta_j^2 - \Delta_t^2) \left(\lambda_e \lambda_o - \frac{1}{4} \right) \bar{g}_{\bar{j}} - 2\bar{\Delta}_t \lambda^+ \bar{g}_{\bar{j}} - \frac{1}{2} \Delta_t^2 \bar{g}_{\bar{j}}, \quad (116b)$$

where the parameters λ^+ and λ^- are assumed to be constants. After the sum and difference of Eqs. (116a) and (116b), we have

$$\bar{g}_j^\pm = \bar{\Delta}_j \bar{f}_j^{\text{eq},\mp} + \bar{\Delta}_t \bar{f}_j^{\text{eq},\pm} - (\Delta_j^2 - \Delta_t^2)(\lambda^+ \mp \lambda^-) \bar{f}_j^{\text{eq},\pm} + (\Delta_j^2 - \Delta_t^2) \left(\lambda_e \lambda_o - \frac{1}{4} \right) \bar{g}_j^\pm - 2\bar{\Delta}_t \lambda^+ \bar{g}_j^\pm - \frac{1}{2} \Delta_t^2 \bar{g}_j^\pm. \quad (117)$$

We also point out that if the terms related to time difference are neglected, then one can also obtain the steady-state REs, which are consistent with the known ones in the previous works [63,64]. Additionally, from Eq. (116a) one can obtain the following zeroth- and first-order moment equations:

$$(\bar{\Delta}_t + \lambda_o \bar{\Delta}_t) \sum_j \bar{f}_j^{\text{eq}} + \sum_j (\bar{\Delta}_j - \lambda_o \Delta_j^2) \bar{f}_j^{\text{eq}} + \left(\lambda_e \lambda_o - \frac{1}{4} \right) \sum_j \Delta_j^2 \bar{g}_j = \Delta t \left[\left(\lambda_e \lambda_o + \frac{1}{4} \right) \Delta_t^2 + 2\lambda^+ \bar{\Delta}_t + 1 \right] \sum_j \bar{F}_j, \quad (118a)$$

$$(\bar{\Delta}_t + \lambda_e \bar{\Delta}_t^2) \sum_j \mathbf{c}_j \bar{f}_j^{\text{eq}} + \sum_j \mathbf{c}_j (\bar{\Delta}_j - \lambda_e \Delta_j^2) \bar{f}_j^{\text{eq}} + \left(\lambda_e \lambda_o - \frac{1}{4} \right) \sum_j \mathbf{c}_j \Delta_j^2 \bar{g}_j = \Delta t \left[\left(\lambda_e \lambda_o + \frac{1}{4} \right) \Delta_t^2 + 2\lambda^+ \bar{\Delta}_t + 1 \right] \sum_j \mathbf{c}_j \bar{F}_j, \quad (118b)$$

where Eqs. (29) and (82a) have been adopted. It should be noted that when $\lambda_e \lambda_o = 1/4$, we can obtain the macroscopic finite-difference scheme of the TRT-LB model, as pointed out in Refs. [63,64]. Actually, for the NCDE, only (118a) is needed, while for the NSEs, we need to consider Eqs. (118a) and (118b).

(3) RLB models [28–30,32]: For simplicity, we first introduce the following tensors:

$$\mathbf{R}_{jk} = \omega_j \frac{\mathbf{c}_j \cdot \mathbf{c}_k}{c_s^2}, \quad \mathbf{P}_{jk} = \omega_j \frac{\mathbf{Q}_j : \mathbf{c}_k \mathbf{c}_k}{2c_s^4}, \quad \mathbf{Q}_j = \mathbf{c}_j \mathbf{c}_j - c_s^2 \mathbf{I}. \quad (119)$$

In the RLB model, the collision matrix $\mathbf{\Lambda}$ is taken as $\mathbf{\Lambda} = \mathbf{I} - (1 - 1/\tau)\mathbf{R}$ for NCDE, while for NSEs, $\mathbf{\Lambda} = \mathbf{I} - (1 - 1/\tau)\mathbf{P}$. It is clear that as a general form, we can take $\mathbf{\Lambda} = \mathbf{I} - (1 - 1/\tau)(\mathbf{R} + \mathbf{P})$ for both NCDE and NSEs.

(4) MLK schemes [25,26]: For the NCDE (44), the collision matrix $\mathbf{\Lambda}$ in the MLK scheme is given by

$$\mathbf{\Lambda} = \frac{1}{\tau} \mathbf{I} + \left(\frac{1}{\tau - A} - \frac{1}{\tau} \right) \mathbf{R}, \quad (120)$$

while for the NSEs (78), it should be taken as

$$\mathbf{\Lambda} = \frac{1}{\tau} \mathbf{I} + \left(\frac{1}{\tau - A} - \frac{1}{\tau} \right) \mathbf{P}. \quad (121)$$

As a general case, the collision matrix $\mathbf{\Lambda}$ can be specified by

$$\mathbf{\Lambda} = \frac{1}{\tau} \mathbf{I} + \left(\frac{1}{\tau - A} - \frac{1}{\tau} \right) (\mathbf{R} + \mathbf{P}), \quad (122)$$

where $\mathbf{R}$ and $\mathbf{P}$ are the same as those in Eq. (119).

(5) Classical MRT-LB models [12–14]: In the classical MRT-LB models, the collision matrix $\mathbf{\Lambda}$ can be determined by

$$\mathbf{\Lambda} = \mathbf{M}^{-1} \mathbf{S} \mathbf{M}, \quad (123)$$

where $\mathbf{S} = \text{diag}(S_0, S_1, S_2, \dots, S_{q-1})$ is a standard diagonal relaxation matrix, $\mathbf{M}$ is the transformation matrix and is composed of the orthogonal or nonorthogonal eigenvectors [4,5,53].

(6) Block triple-relaxation-time LB model [39]: For brevity we first introduce the following $\bar{\mathbf{R}}$ and $\bar{\mathbf{P}}$ in the block triple-relaxation-time LB model:

$$\bar{\mathbf{R}}_{jk} = \omega_j \frac{\mathbf{c}_j \cdot [(\mathbf{S}_1 - S_0 \mathbf{I}) \mathbf{c}_k]}{c_s^2},$$

$$\bar{\mathbf{P}}_{jk} = \omega_j \frac{\mathbf{Q}_j : [(\mathbf{K}_2 - S_0 \hat{\mathbf{I}}) \circ (\mathbf{c}_k \mathbf{c}_k)]}{2c_s^4}, \quad (124)$$

where $\hat{\mathbf{I}}$ is a matrix with $\hat{I}_{ij} = 1$, the symbol $\circ$ represents the Hadamard product, and $\mathbf{K}_2$ is a tensor related to S_2 [39]. Then the collision matrix $\mathbf{\Lambda}$ can be given by

$$\mathbf{\Lambda} = S_0 \mathbf{I} + \bar{\mathbf{R}} + \bar{\mathbf{P}}. \quad (125)$$

(7) Cascaded or central-moment LB model [47]: In the cascaded or central-moment LB model [37,47], we can write the collision matrix as

$$\mathbf{\Lambda} = (\mathbf{N} \mathbf{M})^{-1} \bar{\mathbf{S}} (\mathbf{N} \mathbf{M}) = \mathbf{M}^{-1} \mathbf{S} \mathbf{M}, \quad (126)$$

where $\mathbf{S} = \mathbf{N}^{-1} \bar{\mathbf{S}} \mathbf{N}$ (the MRT-LB models with orthogonal eigenvectors also have this form), $\mathbf{M}$ is the transformation matrix in the MRT-LB model, and $\mathbf{N}$ is a matrix related to macroscopic velocity $\mathbf{u}$ [37,49,50]. Actually, for the collision matrix $\mathbf{\Lambda}$ given by Eq. (126), we can show that the relations (82) and (83) are still satisfied and can be used to derive correct NCDE and NSEs.

(8) Hermite or central-Hermite moment LB model [37]: In the Hermite or central-Hermite moment LB model, the collision matrix can be written as

$$\mathbf{\Lambda} = (\mathbf{B} \mathbf{M})^{-1} \bar{\mathbf{S}} (\mathbf{B} \mathbf{M}) = \mathbf{M}^{-1} \mathbf{S} \mathbf{M}, \quad (127)$$

where $\mathbf{B}$ is a lower-triangular matrix [37].

VI. CONCLUSIONS

In this work, we developed a general framework for the modeling of the MRT-LB method for the NCDE and NSEs that can be used to describe the fluid flows, heat, and mass transfer. The present MRT-LB method with a block-lower-triangular-relaxation matrix can be obtained from the discrete-velocity Boltzmann equation through some proper discretizations of time and space (see Appendix for details). We also conducted a detailed analysis on the CE analysis and MI, DTE, and RE approaches that have been adopted to derive the macroscopic equations from the LB models. The results show that these four analysis methods are equivalent to each other at the second-order equations of expansion parameters and can be used to derive the same macroscopic NCDE and NSEs from present MRT-LB method once the truncation errors are neglected. In addition, we provided some details on the equilibrium, auxiliary, and source distribution functions that should be designed properly to derive the correct NCDE and/or NSEs, and a special discussion on how to develop the local schemes for the diffusion flux (or $\nabla \phi$) and shear stress (or strain rate tensor) is presented. We also found that the existing LB models, including the lattice BGK or SRT-LB model, TRT-LB model, RLB model, MLK schemes, classical MRT-LB model, the block triple-relaxation-time LB model, central-moment LB model, and Hermite and central-Hermite moment LB models, are special cases of the present MRT-LB method, while some recently developed LB models, for instance, the cumulant LB model developed by Geier *et al.* [51], cannot be considered as the particular versions of the MRT-LB model [37]. We also note that although the equilibrium distribution function with second order of discrete velocity is only considered in this work, the cases with high order of discrete velocity can also be adopted in the present framework

of the MRT-LB method. Finally, we point out that the present MRT-LB method can also be used to study the binary mixture problems [86,87], while it cannot be directly applied to the cases with more than two components [45,45,88,89] where the cross diffusion among different components are included, this more complicated problem would be considered in a future work.

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APPENDIX A: FROM DISCRETE-VELOCITY BOLTZMANN EQUATION TO LATTICE BOLTZMANN EQUATION

The MRT-LB method [Eq. (1)] can be obtained from the discrete-velocity Boltzmann equation (DVBE) with invertible

collision matrix $\tilde{\Lambda}$,

$$\begin{aligned} \partial_t f_j(\mathbf{x}, t) + \mathbf{c}_j \cdot \nabla f_j(\mathbf{x}, t) \\ = -\tilde{\Lambda}_{jk} f_k^{\text{ne}}(\mathbf{x}, t) + \tilde{G}_j(\mathbf{x}, t) + F_j(\mathbf{x}, t), \end{aligned} \quad (\text{A1})$$

which can be also considered as a basic equation to develop some other mesoscopic methods (e.g., the discrete-unified gas-kinetic scheme [90,91]).

Compared to Eq. (1), the source distribution function in DVBE (A1) is the sum of F_j and $\tilde{G}_j$, F_j is used as the source or forcing term, and $\tilde{G}_j$ is adopted to remove some additional terms.

Integrating Eq. (A1) along the characteristic line $\mathbf{x}' = \mathbf{x} + \mathbf{c}_j t'$ with $t' \in [0, \Delta t]$, we obtain

$$\begin{aligned} f_j(\mathbf{x} + \mathbf{c}_j \Delta t, t + \Delta t) \\ = f_j(\mathbf{x}, t) + \int_0^{\Delta t} (-\tilde{\Lambda}_{jk} f_k^{\text{ne}} + \tilde{G}_j + F_j)(\mathbf{x} + \mathbf{c}_j t', t + t') dt', \end{aligned} \quad (\text{A2})$$

Based on the following results:

$$\int_0^{\Delta t} (-\tilde{\Lambda}_{jk} f_k^{\text{ne}} + \tilde{G}_j)(\mathbf{x} + \mathbf{c}_j t', t + t') dt' = \frac{\Delta t}{2} \left[(-\tilde{\Lambda}_{jk} f_k^{\text{ne}} + \tilde{G}_j)(\mathbf{x} + \mathbf{c}_j \Delta t, t + \Delta t) + (-\tilde{\Lambda}_{jk} f_k^{\text{ne}} + \tilde{G}_j)(\mathbf{x}, t) \right] + O(\Delta t^3), \quad (\text{A3a})$$

$$\begin{aligned} \int_0^{\Delta t} F_j(\mathbf{x} + \mathbf{c}_j t', t + t') dt' &= \int_0^{\Delta t} [F_j(\mathbf{x}, t) + t' D_j F_j(\mathbf{x}, t) + O(t'^2)] dt' \\ &= \Delta t \left[F_j(\mathbf{x}, t) + \frac{\Delta t}{2} D_j F_j(\mathbf{x}, t) \right] + O(\Delta t^3), \end{aligned} \quad (\text{A3b})$$

we can write Eq. (A2) as

$$\tilde{f}_j(\mathbf{x} + \mathbf{c}_j \Delta t, t + \Delta t) = \tilde{f}_j(\mathbf{x}, t) + O(\Delta t^3), \quad (\text{A4})$$

where

$$\tilde{f}_j = f_j - \frac{\Delta t}{2} (-\tilde{\Lambda}_{jk} f_k^{\text{ne}} + \tilde{G}_j), \quad (\text{A5a})$$

$$\tilde{f}_j = f_j + \frac{\Delta t}{2} (-\tilde{\Lambda}_{jk} f_k^{\text{ne}} + \tilde{G}_j) + \Delta t \left(F_j + \frac{\Delta t}{2} D_j F_j \right). \quad (\text{A5b})$$

Actually, Eq. (A5) can also be written as

$$\tilde{\mathbf{f}} = \mathbf{f} - \frac{\Delta t}{2} (-\tilde{\Lambda} \mathbf{f}^{\text{ne}} + \tilde{\mathbf{G}}) = \left(\mathbf{I} + \frac{\Delta t}{2} \tilde{\Lambda} \right) \mathbf{f} - \frac{\Delta t}{2} (\tilde{\Lambda} \mathbf{f}^{\text{eq}} + \tilde{\mathbf{G}}), \quad (\text{A6a})$$

$$\tilde{\mathbf{f}} = \left(\mathbf{I} - \frac{\Delta t}{2} \tilde{\Lambda} \right) \mathbf{f} + \frac{\Delta t}{2} (\tilde{\Lambda} \mathbf{f}^{\text{eq}} + \tilde{\mathbf{G}}) + \Delta t \left(\mathbf{F} + \frac{\Delta t}{2} \mathbf{D} \mathbf{F} \right). \quad (\text{A6b})$$

From the above equation, we have

$$\mathbf{f} = \left(\mathbf{I} + \frac{\Delta t}{2} \tilde{\Lambda} \right)^{-1} \left[\tilde{\mathbf{f}} + \frac{\Delta t}{2} (\tilde{\Lambda} \mathbf{f}^{\text{eq}} + \tilde{\mathbf{G}}) \right], \quad (\text{A7a})$$

$$\begin{aligned} \tilde{\mathbf{f}} &= \left(\mathbf{I} - \frac{\Delta t}{2} \tilde{\Lambda} \right) \left(\mathbf{I} + \frac{\Delta t}{2} \tilde{\Lambda} \right)^{-1} \left[\tilde{\mathbf{f}} + \frac{\Delta t}{2} (\tilde{\Lambda} \mathbf{f}^{\text{eq}} + \tilde{\mathbf{G}}) \right] \\ &\quad + \frac{\Delta t}{2} (\tilde{\Lambda} \mathbf{f}^{\text{eq}} + \tilde{\mathbf{G}}) + \Delta t \left(\mathbf{F} + \frac{\Delta t}{2} \mathbf{D} \mathbf{F} \right). \end{aligned} \quad (\text{A7b})$$

Let

$$\mathbf{I} - \Lambda = \left(\mathbf{I} - \frac{\Delta t}{2} \tilde{\Lambda} \right) \left(\mathbf{I} + \frac{\Delta t}{2} \tilde{\Lambda} \right)^{-1}, \quad (\text{A8})$$

then

$$\begin{aligned} \tilde{\mathbf{f}} &= (\mathbf{I} - \Lambda) \tilde{\mathbf{f}} + [\mathbf{I} + (\mathbf{I} - \Lambda)] \left[\frac{\Delta t}{2} (\tilde{\Lambda} \mathbf{f}^{\text{eq}} + \tilde{\mathbf{G}}) \right] \\ &\quad + \Delta t \left[\mathbf{F} + \frac{\Delta t}{2} \mathbf{D} \mathbf{F} \right]. \end{aligned} \quad (\text{A9})$$

According to the following equation:

$$\begin{aligned}
 [\mathbf{I} + (\mathbf{I} - \mathbf{\Lambda})] \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} &= \left[\mathbf{I} + \left(\mathbf{I} - \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right) \left(\mathbf{I} + \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right)^{-1} \right] \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \\
 &= \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} + \left(\mathbf{I} - \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right) \left(\mathbf{I} + \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right)^{-1} \left(\mathbf{I} + \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} - \mathbf{I} \right) \\
 &= \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} + \left(\mathbf{I} - \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right) - \left(\mathbf{I} - \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right) \left(\mathbf{I} + \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right)^{-1} = \mathbf{\Lambda},
 \end{aligned} \tag{A10}$$

where Eq. (A8) has been used, we can write Eq. (A9) as

$$\begin{aligned}
 \tilde{\mathbf{f}} &= (\mathbf{I} - \mathbf{\Lambda}) \tilde{\mathbf{f}} + \mathbf{\Lambda} \mathbf{f}^{\text{eq}} + \Delta t \left(\mathbf{I} - \frac{1}{2} \mathbf{\Lambda} \right) \tilde{\mathbf{G}} + \Delta t \left(\mathbf{F} + \frac{\Delta t}{2} \mathbf{D} \mathbf{F} \right) \\
 &= \tilde{\mathbf{f}} - \mathbf{\Lambda} (\tilde{\mathbf{f}} - \mathbf{f}^{\text{eq}}) + \Delta t \left[\left(\mathbf{I} - \frac{1}{2} \mathbf{\Lambda} \right) \tilde{\mathbf{G}} + \mathbf{F} + \frac{\Delta t}{2} \mathbf{D} \mathbf{F} \right].
 \end{aligned} \tag{A11}$$

Substituting above equation into Eq. (A4) and removing the term $O(\Delta t^3)$, we obtain

$$\begin{aligned}
 \tilde{f}_j(\mathbf{x} + \mathbf{c}_j \Delta t, t + \Delta t) &= \tilde{f}_j - \mathbf{\Lambda}_{jk} (\tilde{f}_k - f_k^{\text{eq}}) \\
 &\quad + \Delta t \left(G_j + F_j + \frac{\Delta t}{2} D_j F_j \right),
 \end{aligned} \tag{A12}$$

where $\mathbf{G} = (\mathbf{I} - \mathbf{\Lambda}/2) \tilde{\mathbf{G}}$, the relaxation matrix $\mathbf{\Lambda}$ is given by

$$\begin{aligned}
 \mathbf{\Lambda} &= \mathbf{I} - \left(\mathbf{I} - \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right) \left(\mathbf{I} + \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right)^{-1} \\
 &= \left(\mathbf{I} + \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right) \left(\mathbf{I} + \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right)^{-1} \\
 &\quad - \left(\mathbf{I} - \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right) \left(\mathbf{I} + \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right)^{-1} \\
 &= \Delta t \tilde{\mathbf{\Lambda}} \left(\mathbf{I} + \frac{\Delta t}{2} \tilde{\mathbf{\Lambda}} \right)^{-1} \\
 &= \left(\frac{1}{2} \mathbf{I} + \frac{1}{\Delta t} \tilde{\mathbf{\Lambda}}^{-1} \right)^{-1}.
 \end{aligned} \tag{A13}$$

We note that Eq. (A12) has the same form as Eq. (1) except the last term on the right-hand side. Actually, if $\gamma = 1$, Eq. (1) would reduce to Eq. (A12), while $\gamma = 0$ can also be used derive correct NCDE [17].

APPENDIX B: FROM DISCRETE-VELOCITY BOLTZMANN EQUATION TO THE MACROSCOPIC EQUATIONS THROUGH CE METHOD

It is known from the above discussion that two classes of macroscopic equations, i.e., NCDE and NSEs, can be recovered correctly from MRT-LB method (1) through some analysis methods (CE analysis and the MI, DTE, and RE methods) when the moments of the distribution functions f_j^{eq} , G_j , and F_j are given properly. Here only the CE analysis is considered to obtain the NCDE and NSEs.

Based on the CE analysis, we have

$$\begin{aligned}
 f_j &= f_j = f_j^{(0)} + \epsilon f_j^{(1)} + \epsilon^2 f_j^{(2)}, \quad \tilde{G}_j = \epsilon \tilde{G}_j^{(1)} + \epsilon^2 \tilde{G}_j^{(2)}, \\
 F_j &= \epsilon F_j^{(1)} + \epsilon^2 F_j^{(2)},
 \end{aligned} \tag{B1a}$$

$$\partial_t = \epsilon \partial_{t_1} + \epsilon^2 \partial_{t_2}, \quad \nabla = \epsilon \nabla_1, \quad D_j = \epsilon D_{1j} + \epsilon^2 \partial_{t_2}. \tag{B1b}$$

Substituting Eq. (B1) into Eq. (A1), we can obtain the following equations at the order of ϵ^0 , ϵ^1 and ϵ^2 :

$$f_j^{(0)} - f_j^{\text{eq}} = 0, \tag{B2a}$$

$$D_{1j} f_j^{(0)} = -\tilde{\mathbf{\Lambda}}_{jl} f_l^{(1)} + \tilde{G}_j^{(1)} + F_j^{(1)}, \tag{B2b}$$

$$\partial_{t_2} f_j^{(0)} + D_{1j} f_j^{(1)} = -\tilde{\mathbf{\Lambda}}_{jl} f_l^{(2)} + \tilde{G}_j^{(2)} + F_j^{(2)}. \tag{B2c}$$

It can be seen that Eq. (B2) derived from DVBE (A1) is simpler than Eq. (5) obtained from MRT-LB method (1) in which some discrete terms [e.g., $\Delta t^2 D_{1j} f_j^{(0)}/2$ in Eq. (5c)] caused by the numerical scheme are included. In the following analysis, the form of relaxation matrix defined by Eq. (103) is adopted for simplicity.

1. From DVBE to NCDE through CE analysis

To correctly recover NCDE (44) from the DVBE (A1), the unknown conserved scalar ϕ , f_j , and f_j^{eq} satisfy Eq. (45), while moments of $\tilde{G}_j$, F_j , and $\tilde{\mathbf{\Lambda}}$ are given by

$$\sum_j \tilde{G}_j = 0, \quad \sum_j \mathbf{c}_j \tilde{G}_j = \mathbf{M}_{1\tilde{G}}, \tag{B3a}$$

$$\sum_j F_j = S, \quad \sum_j \mathbf{c}_j F_j = \mathbf{M}_{1F} = 0, \tag{B3b}$$

$$\sum_j \mathbf{e}_j \tilde{\mathbf{\Lambda}}_{jk} = \tilde{s}_0 \mathbf{e}_k, \quad \sum_j \mathbf{c}_j \tilde{\mathbf{\Lambda}}_{jk} = \tilde{\mathbf{S}}_1 \mathbf{c}_k, \quad \forall k, \tag{B3c}$$

where the moment $\mathbf{M}_{1\tilde{G}}$ to be determined below. $\tilde{\mathbf{S}}_1$ is an invertible $d \times d$ relaxation matrix corresponding to the diffusion matrix $\mathbf{K}$.

Similarly to Eq. (47), it follows from Eqs. (45), (B1), (B2a), and (B3) that

$$\sum_j f_j^{(k)} = 0, \tag{B4a}$$

$$\sum_j \tilde{G}_j^{(k)} = 0, \quad \sum_j \mathbf{c}_j \tilde{G}_j^{(k)} = \mathbf{M}_{1\tilde{G}}^{(k)}, \tag{B4b}$$

$$\sum_j F_j^{(k)} = S^{(k)}, \quad \sum_j \mathbf{c}_j F_j^{(k)} = \mathbf{0}, \tag{B4c}$$

where $k \geq 1$. Summing Eq. (B2b) and Eq. (B2c) over j , one can obtain

$$\partial_{t_1} \phi + \nabla_1 \cdot \mathbf{B} = -\tilde{s}_0 \sum_l f_l^{(1)} + S^{(1)} = S^{(1)}, \quad (\text{B5a})$$

$$\partial_{t_2} \phi + \nabla_1 \cdot \sum_l \mathbf{c}_l f_l^{(1)} = S^{(2)}. \quad (\text{B5b})$$

where Eqs. (45), (B3), and (B4) have been used.

Multiplying $\mathbf{c}_j$ on both sides of Eqs. (B2b) and (B2c), and summing them over j , we have

$$\partial_{t_1} \mathbf{B} + \nabla_1 \cdot (\mathbf{C} + \beta c_s^2 \mathbf{D}) = -\tilde{\mathbf{S}}_1 \sum_l \mathbf{c}_l f_l^{(1)} + \mathbf{M}_{1\tilde{G}}, \quad (\text{B6a})$$

$$\begin{aligned} \partial_{t_2} \mathbf{B} + \partial_{t_1} \sum_l \mathbf{c}_l f_l^{(1)} + \nabla_1 \cdot \sum_l \mathbf{c}_l \mathbf{c}_l f_l^{(1)} \\ = -\tilde{\mathbf{S}}_1 \sum_l \mathbf{c}_l f_l^{(2)} + \mathbf{M}_{1\tilde{G}}^{(2)}, \end{aligned} \quad (\text{B6b})$$

where Eqs. (B3) and (B4) have been adopted to derive above equation.

Substituting Eq. (B6a) into Eq. (B5b) gives

$$\partial_{t_2} \phi = \nabla_1 \cdot [\mathbf{K} \cdot (\nabla_1 \cdot \mathbf{D})] + \nabla_1 \cdot RH_1 + S^{(2)}, \quad (\text{B7})$$

where $\mathbf{K} = \beta c_s^2 \tilde{\mathbf{S}}_1^{-1}$ with $\tilde{\mathbf{S}}_1^{-1} = (\mathbf{S}_1^{-1} - \frac{1}{2} \mathbf{I}) \Delta t$, $RH_1 = \tilde{\mathbf{S}}_1^{-1} (\partial_{t_1} \mathbf{B} + \nabla_1 \cdot \mathbf{C} - \mathbf{M}_{1\tilde{G}}^{(1)})$. Taking $\mathbf{M}_{1\tilde{G}}$ such that

$$\partial_t \mathbf{B} + \nabla \cdot \mathbf{C} - \mathbf{M}_{1\tilde{G}} = 0, \quad (\text{B8})$$

we have $RH_1 = 0$, which leads to

$$\partial_{t_2} \phi = \nabla_1 \cdot [\mathbf{K} \cdot (\nabla_1 \cdot \mathbf{D})] + S^{(2)}. \quad (\text{B9})$$

Combining Eq. (B6a) with Eq. (B9) yields the NCDE (44),

$$\partial_t \phi + \nabla \cdot \mathbf{B} = \nabla \cdot [\mathbf{K} \cdot (\nabla \cdot \mathbf{D})] + S. \quad (\text{B10})$$

In addition, it should be noted that from Eq. (B8) and the relation $\mathbf{G} = (\mathbf{I} - \Lambda/2) \tilde{\mathbf{G}}$, we can determine $\mathbf{M}_{1G}$ as

$$\mathbf{M}_{1G} = (\mathbf{I} - \mathbf{S}_1/2) \mathbf{M}_{1\tilde{G}} = (\mathbf{I} - \mathbf{S}_1/2) (\partial_t \mathbf{B} + \nabla \cdot \mathbf{C}), \quad (\text{B11})$$

which is the same as Eq. (57).

2. From DVBE to NSEs through CE analysis

Similar to the above discussion, to recover NSEs (78) from the DVBE (A1), the macroscopic variables ρ and $\mathbf{u}$ and the distribution functions f_j and f_j^{eq} satisfy Eqs. (80a) and (80b), while the moments of $\tilde{G}_j$, F_j , and $\tilde{\Lambda}$ should satisfy the following relations:

$$\sum_j \tilde{G}_j = 0, \quad \sum_j \mathbf{c}_j \tilde{G}_j = \mathbf{M}_{1\tilde{G}} = \mathbf{0}, \quad \sum_j \mathbf{c}_j \mathbf{c}_j \tilde{G}_j = \mathbf{M}_{2\tilde{G}}, \quad (\text{B12a})$$

$$\sum_j F_j = \bar{S}, \quad \sum_j \mathbf{c}_j F_j = \mathbf{M}_{1F} = \bar{\mathbf{F}}, \quad \sum_j \mathbf{c}_j \mathbf{c}_j F_j = \mathbf{M}_{2F} = \mathbf{0}, \quad (\text{B12b})$$

$$\sum_j \mathbf{e}_j \tilde{\Lambda}_{jk} = \tilde{s}_0 \mathbf{e}_k, \quad \sum_j \mathbf{c}_j \tilde{\Lambda}_{jk} = \tilde{\mathbf{S}}_1 \mathbf{c}_k, \quad (\text{B12c})$$

$$\begin{aligned} \sum_j \mathbf{c}_{j\alpha} \mathbf{c}_{j\beta} \tilde{\Lambda}_{jk} &= \sum_j \left(\mathbf{c}_{j\alpha} \mathbf{c}_{j\beta} - \frac{\delta_{\alpha\beta}}{d} \mathbf{c}_{j\gamma} \mathbf{c}_{j\gamma} \right) \tilde{\Lambda}_{jk} + \sum_j \frac{\delta_{\alpha\beta}}{d} \mathbf{c}_{j\gamma} \mathbf{c}_{j\gamma} \tilde{\Lambda}_{jk} \\ &= \tilde{S}_{2s} \left(\mathbf{c}_{k\alpha} \mathbf{c}_{k\beta} - \frac{\delta_{\alpha\beta}}{d} \mathbf{c}_{k\gamma} \mathbf{c}_{k\gamma} \right) + \tilde{S}_{2b} \frac{\delta_{\alpha\beta}}{d} \mathbf{c}_{k\gamma} \mathbf{c}_{k\gamma}, \end{aligned} \quad (\text{B12d})$$

where the moment $\mathbf{M}_{2\tilde{G}}$ will be determined later. $\tilde{S}_{2s}$ and $\tilde{S}_{2b}$ are two relaxation parameters related to dynamic viscosity μ and bulk viscosity μ_b .

Similarly, from Eqs. (80), (B1), (B2a), and (B12) we have

$$\sum_j f_j^{(k)} = 0, \quad \sum_j \mathbf{c}_j f_j^{(k)} = \mathbf{0}, \quad (\text{B13a})$$

$$\sum_j \tilde{G}_j^{(k)} = 0, \quad \sum_j \mathbf{c}_j \tilde{G}_j^{(k)} = \mathbf{0}, \quad \sum_j \mathbf{c}_j \mathbf{c}_j \tilde{G}_j^{(k)} = \mathbf{M}_{2\tilde{G}}^{(k)}, \quad (\text{B13b})$$

$$\sum_j F_j^{(k)} = \bar{S}^{(k)}, \quad \sum_j \mathbf{c}_j F_j^{(k)} = \bar{\mathbf{F}}^{(k)}, \quad \sum_j \mathbf{c}_j \mathbf{c}_j F_j^{(k)} = \mathbf{0}, \quad (\text{B13c})$$

where $k \geq 1$. Summing Eqs. (B2b) and (B2c) over j , and using Eqs. (80a), (B12), and (B13), we can obtain

$$\partial_{t_1} \rho + \nabla_1 \cdot (\rho \mathbf{u}) = -\tilde{s}_0 \sum_l f_l^{(1)} + \bar{S}^{(1)} = \bar{S}^{(1)}, \quad (\text{B14a})$$

$$\partial_{t_2} \rho = -\tilde{s}_0 \sum_l f_l^{(2)} + \bar{S}^{(2)} = \bar{S}^{(2)}. \quad (\text{B14b})$$

Multiplying $\mathbf{c}_j$ and $\mathbf{c}_j \mathbf{c}_j$ on both sides of Eqs. (B2b) and (B2c), and summing these equations over j , one can obtain

$$\begin{aligned} \partial_{t_1} (\rho \mathbf{u}) + \nabla_1 \cdot (c_s^2 \rho \mathbf{I} + \rho \mathbf{u} \mathbf{u}) &= -\tilde{\mathbf{S}}_1 \sum_l \mathbf{c}_l f_l^{(1)} + \mathbf{M}_{1F}^{(1)} \\ &= \bar{\mathbf{F}}^{(1)}, \end{aligned} \quad (\text{B15a})$$

$$\partial_{t_2} (\rho \mathbf{u}) + \nabla_1 \cdot \sum_l \mathbf{c}_l \mathbf{c}_l f_l^{(1)} = -\tilde{\mathbf{S}}_1 \sum_l \mathbf{c}_l f_l^{(2)} + \mathbf{M}_{1F}^{(2)} = \bar{\mathbf{F}}^{(2)}, \quad (\text{B15b})$$

$$\partial_{t_1}(c_s^2 \rho \mathbf{I} + \rho \mathbf{u}\mathbf{u}) + \nabla_1 \cdot [c_s^2 \Delta \cdot (\rho \mathbf{u})] = -\tilde{S}_{2s} \sum_l \left(\mathbf{c}_l \mathbf{c}_l - \frac{1}{d} \mathbf{c}_l \cdot \mathbf{c}_l \mathbf{I} \right) f_l^{(1)} - \tilde{S}_{2b} \sum_l \left(\frac{1}{d} \mathbf{c}_l \cdot \mathbf{c}_l \mathbf{I} \right) f_l^{(1)} + \mathbf{M}_{2\tilde{G}}^{(1)}, \quad (\text{B16a})$$

$$\begin{aligned} & \partial_{t_2}(c_s^2 \rho \mathbf{I} + \rho \mathbf{u}\mathbf{u}) + \partial_{t_1} \sum_l \mathbf{c}_l \mathbf{c}_l f_l^{(1)} + \nabla_1 \cdot \sum_l \mathbf{c}_l \mathbf{c}_l f_l^{(1)} \\ &= -\tilde{S}_{2s} \sum_l \left(\mathbf{c}_l \mathbf{c}_l - \frac{1}{d} \mathbf{c}_l \cdot \mathbf{c}_l \mathbf{I} \right) f_l^{(2)} - \tilde{S}_{2b} \sum_l \left(\frac{1}{d} \mathbf{c}_l \cdot \mathbf{c}_l \mathbf{I} \right) f_l^{(2)} + \mathbf{M}_{2\tilde{G}}^{(2)}, \end{aligned} \quad (\text{B16b})$$

where Eqs. (B12) and (B13) have been used.

From Eq. (B14), the continuity equation (78a) is **recovered correctly to the order of $O(\epsilon^2)$** ,

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = \tilde{S}. \quad (\text{B17})$$

On the other hand, with the help of Eq. (B16), we can obtain the momentum equation (78b) from Eq. (B15). According to Eqs. (B12a) and (B15a), we can rewrite Eq. (B16a) as

$$\begin{aligned} \partial_{t_1}(c_s^2 \rho \mathbf{I} + \rho \mathbf{u}\mathbf{u}) + \nabla_1 \cdot [c_s^2 \Delta \cdot (\rho \mathbf{u})] &= c_s^2 \tilde{S}^{(1)} \mathbf{I} + \mathbf{u} \tilde{\mathbf{F}}^{(1)} + \tilde{\mathbf{F}}^{(1)} \mathbf{u} + \mathbf{u}\mathbf{u} \tilde{S}^{(1)} + \rho c_s^2 [\nabla_1 \mathbf{u} + (\nabla_1 \mathbf{u})^T] \\ &= -\tilde{S}_{2s} \sum_l \left(\mathbf{c}_l \mathbf{c}_l - \frac{1}{d} \mathbf{c}_l \cdot \mathbf{c}_l \mathbf{I} \right) f_l^{(1)} - \tilde{S}_{2b} \sum_l \left(\frac{1}{d} \mathbf{c}_l \cdot \mathbf{c}_l \mathbf{I} \right) f_l^{(1)} + \mathbf{M}_{2\tilde{G}}^{(1)}. \end{aligned} \quad (\text{B18})$$

To derive correct momentum equation, the following conditions should be satisfied:

$$\rho c_s^2 [\nabla_1 \mathbf{u} + (\nabla_1 \mathbf{u})^T] = -\tilde{S}_{2s} \sum_l \left(\mathbf{c}_l \mathbf{c}_l - \frac{1}{d} \mathbf{c}_l \cdot \mathbf{c}_l \mathbf{I} \right) f_l^{(1)} - \tilde{S}_{2b} \sum_l \left(\frac{1}{d} \mathbf{c}_l \cdot \mathbf{c}_l \mathbf{I} \right) f_l^{(1)}, \quad (\text{B19a})$$

$$\mathbf{M}_{2\tilde{G}}^{(1)} = c_s^2 \tilde{S}^{(1)} \mathbf{I} + \mathbf{u} \tilde{\mathbf{F}}^{(1)} + \tilde{\mathbf{F}}^{(1)} \mathbf{u} + \mathbf{u}\mathbf{u} \tilde{S}^{(1)}. \quad (\text{B19b})$$

Then from Eq. (B19), we can obtain

$$\sum_l \mathbf{c}_l \mathbf{c}_l f_l^{(1)} = -\rho c_s^2 \left\{ \frac{1}{\tilde{S}_{2s}} \left[\nabla_1 \mathbf{u} + (\nabla_1 \mathbf{u})^T - \frac{2}{d} (\nabla_1 \cdot \mathbf{u}) \mathbf{I} \right] + \frac{1}{\tilde{S}_{2b}} \frac{2}{d} (\nabla_1 \cdot \mathbf{u}) \mathbf{I} \right\}. \quad (\text{B20})$$

Substituting Eqs. (B20) into Eq. (B15b) yields the following equation:

$$\partial_{t_2}(\rho \mathbf{u}) = \rho c_s^2 \left\{ \frac{1}{\tilde{S}_{2s}} \left[\nabla_1 \mathbf{u} + (\nabla_1 \mathbf{u})^T - \frac{2}{d} (\nabla_1 \cdot \mathbf{u}) \mathbf{I} \right] + \frac{1}{\tilde{S}_{2b}} \frac{2}{d} (\nabla_1 \cdot \mathbf{u}) \mathbf{I} \right\} + \tilde{\mathbf{F}}^{(2)}. \quad (\text{B21})$$

Combining Eq. (B15a) with Eq. (B21) we have

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla P + \nabla \cdot \mu \left[\nabla \mathbf{u} + (\nabla \mathbf{u})^T - \frac{2}{d} (\nabla \cdot \mathbf{u}) \mathbf{I} \right] + \nabla(\mu_b \nabla \cdot \mathbf{u}) + \tilde{\mathbf{F}}, \quad (\text{B22})$$

where the pressure P and the dynamic and bulk viscosities are given by

$$P = \rho c_s^2, \quad \mu = \rho c_s^2 / \tilde{S}_{2s}, \quad \mu_b = 2\rho c_s^2 / (\tilde{S}_{2b} d). \quad (\text{B23})$$

From Eqs. (B17) and (B22), it is clear that through the CE analysis, the NSEs (78) can be correctly recovered from the DVBE (A1) at the order of $O(\epsilon^2)$. In addition, we also point out that the relaxation parameters $\tilde{S}_{2s}$ and $\tilde{S}_{2b}$ in the DVBE are related to S_{2s} and S_{2b} in the MRT-LB method through $\tilde{S}_{2s}^{-1} = (S_{2s}^{-1} - 1/2)\Delta t$ and $\tilde{S}_{2b}^{-1} = (S_{2b}^{-1} - 1/2)\Delta t$.

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